

TurboWin+ version 4.2 Noteworthy features and Tips

- 1 Email options
- 2 Vaisala HMP155 sensor interface
- 3 EMOS (Mintaka) interface
- 4 Setting TurboWin+ to APR (or APTR)
- 5 Station ID
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- 7 Barometer comparison form (barometer self-check form)
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1 Email options

1.1] General

1.2] Setting up local host for TurboWin+

1.3] Setting up Gmail for TurboWin+

1.4] Setting up Yahoo Mail for TurboWin+

1.5] Sending and Logging

1.1] General

Two settings (address recipient and subject) are always required before you can send observations by TurboWin+ (Maintenance → Email settings).

E-mail settings

for 'Output > Obs by Email' and AP[&T]R / AWSR

[ALL] address recipient cc* Email subject

[FORMAT 101]** ☐ obs in body ☒ obs in attachment

[CUSTOM] your email address server

password port ☒ TLS ☐ SSL ☒ STARTTLS

* optional ** consult your PMO

for 'Maintenance > Move log files by Email'

logs email recipient These log files include important data which is of particular value for climate studies
Downloading of the log files should be done at routine intervals (ideally not exceeding 6 months)

OK Cancel

fig 1 Email settings example with required basic data surrounded in red

1.2] Setting up (SMTP) local host for TurboWin+ Custom email

Sending via local host [Output – Obs by Email (Custom)] could be an alternative for sending via the default email app. Ask the system administrator for the required settings.

1.3] Setting up Gmail for TurboWin+ Custom email

basically this are the required steps

1. If no account create a new account [<https://support.google.com/mail/answer/56256?hl=en>]
2. For this account turn on 2-step verification [<https://support.google.com/accounts/answer/185839>]
3. Generate an app password [e.g. <https://www.lifewire.com/get-a-password-to-access-gmail-by-pop-imap-2-1171882>] [see note 1 below and fig 3]
4. Insert your Gmail address into the TurboWin+ Email settings form [fig 4]
5. Insert the Gmail server name into the TurboWin+ Email settings form [fig 4]
6. Paste the (app) password into the (app) password field in the TurboWin+ Email settings form [see note 2 below and fig 4]
7. Insert the appropriate port number into the TurboWin+ Email settings form [see note 3 below and fig 4]
8. Select the appropriate security settings (TLS, SSL, STARTTLS) in the TurboWin+ Email settings form [see note 3 below and fig 4]

Note 1

The value of an application-specific password is that you can revoke and regenerate a password on a service-by-service basis instead of having to change the master password to your account. If you do need to create a new app-specific password for a program or service, revoke passwords previously set up but no longer used for the same application.

It's best practice to use an app-specific password only for a single service. You are free to generate as many app-specific passwords as you like. [source: <https://www.lifewire.com/get-a-password-to-access-gmail-by-pop-imap-2-1171882>]

Note 2

Most of the time, you'll only have to enter an (app) password once per app or device, so don't worry about memorizing it. You can also always generate a new one.

Note 3

TLS uses port 587 and SSL port 465. One or both ports could be initially closed. TLS is recommended. Ask the system administrator

← App passwords

App passwords let you sign in to your Google Account from apps on devices that don't support 2-Step Verification. You'll only need to enter it once so you don't need to remember it. [Learn more](#)

Your app passwords

Name	Created	Last used	
pc	Jul 17	1:39 PM	

Select the app and device you want to generate the app password for.

Select app ▼

Select device ▼

GENERATE

fig 3 View of app passwords on Google account web site for (test)email address turbowin.observations@gmail.com

E-mail settings

for 'Output > Obs by Email' and AP[&T]R / AWSR

[ALL] address recipient cc* Email subject

[FORMAT 101]** ☐ obs in body ☒ obs in attachment

[CUSTOM] your email address server

password port ☒ TLS ☐ SSL ☒ STARTTLS

* optional ** consult your PMO

for 'Maintenance > Move log files by Email'

logs email recipient

These log files include important data which is of particular value for climate studies
Downloading of the log files should be done at routine intervals (ideally not exceeding 6 months)

OK Cancel

fig 4 example, setting up Gmail for TurboWin+

1.4] Setting up Yahoo Mail for TurboWin+ Custom Email

basically this are the required steps

1. If no account create a new account [<https://login.yahoo.com/account/create?.lang=en-US&.intl=us&.src=yhelp>]
2. For this account turn on 2-step verification [<https://help.yahoo.com/kb/SLN5013.html>]
3. Generate an app password [<https://uk.help.yahoo.com/kb/account/generate-third-party-passwords-sln15241.html>] [see note 1 below and fig 6 and fig 7]
4. Insert your Yahoo address into the TurboWin+ Email settings form [fig 8]
5. Insert the Yahoo server name into the TurboWin+ Email settings form [fig 8]
6. Paste the (app) password into the (app) password field in the TurboWin+ Email settings form [see note 2 below and fig 8]
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8. Select the appropriate security settings (TLS, SSL, STARTTLS) in the TurboWin+ Email settings form [see note 3 below and fig 8]

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TLS uses port 587 and SSL port 465. One or both ports could be initially closed. TLS is recommended. Ask the system administrator

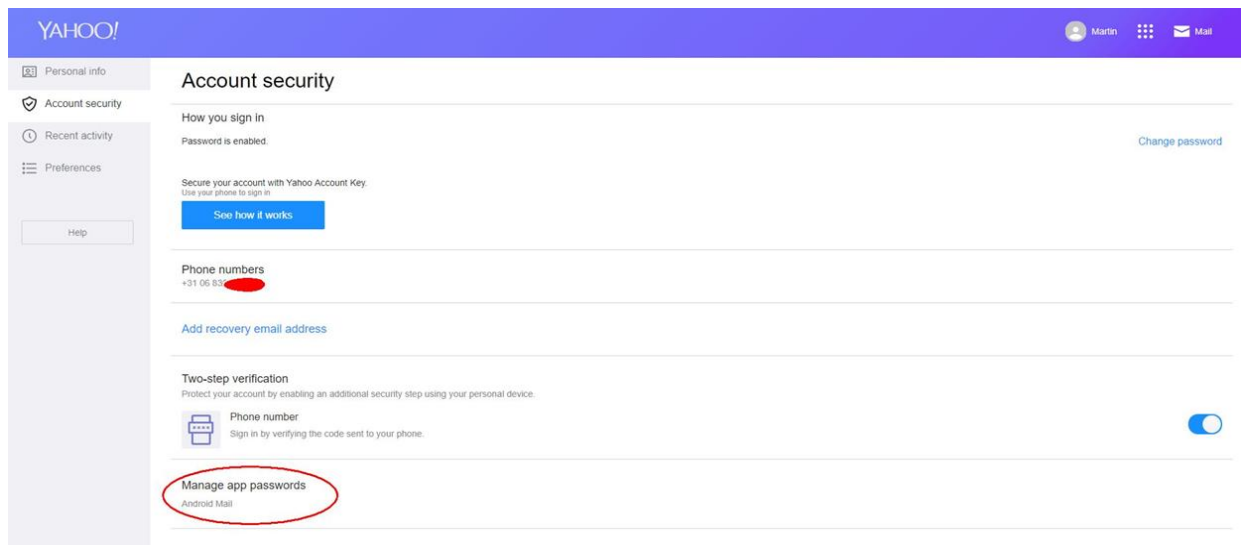


fig 6 setting up Yahoo email for TurboWin+ (Yahoo account web site)

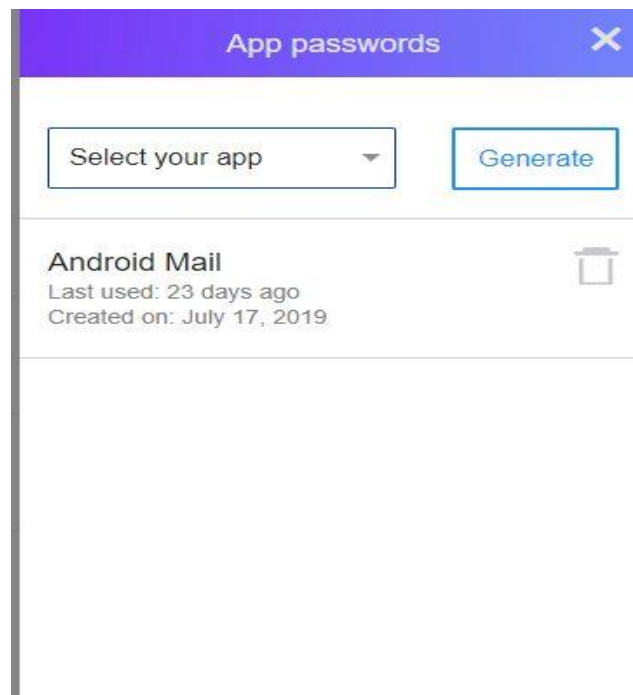


fig 7 View of app passwords on Yahoo account web site for (test)email address turbowin@yahoo.com

E-mail settings

for 'Output > Obs by Email' and AP[&T]R / AWSR

[ALL] address recipient cc* Email subject

[FORMAT 101]** ☐ obs in body ☒ obs in attachment

[CUSTOM] your email address server
password port ☒ TLS ☐ SSL ☒ STARTTLS

* optional ** consult your PMO

for 'Maintenance > Move log files by Email'

logs email recipient

These log files include important data which is of particular value for climate studies
Downloading of the log files should be done at routine intervals (ideally not exceeding 6 months)

OK Cancel

fig 8 example, setting up Yahoo Mail for TurboWin+

1.5 Sending and Logging

In manual mode: an Email output option (fig 10) will only be available (not grayed out) if the required settings for that specific option are made before in the Email settings form (fig 1).

note

In APR, APTR and AWSR mode all output options are not available (grayed out) by default. If you turn off APR (APTR or AWSR) the appropriate output options will be selectable again. In EUCAWS mode only the output option “Obs to AWS” is available for adding manual observation data

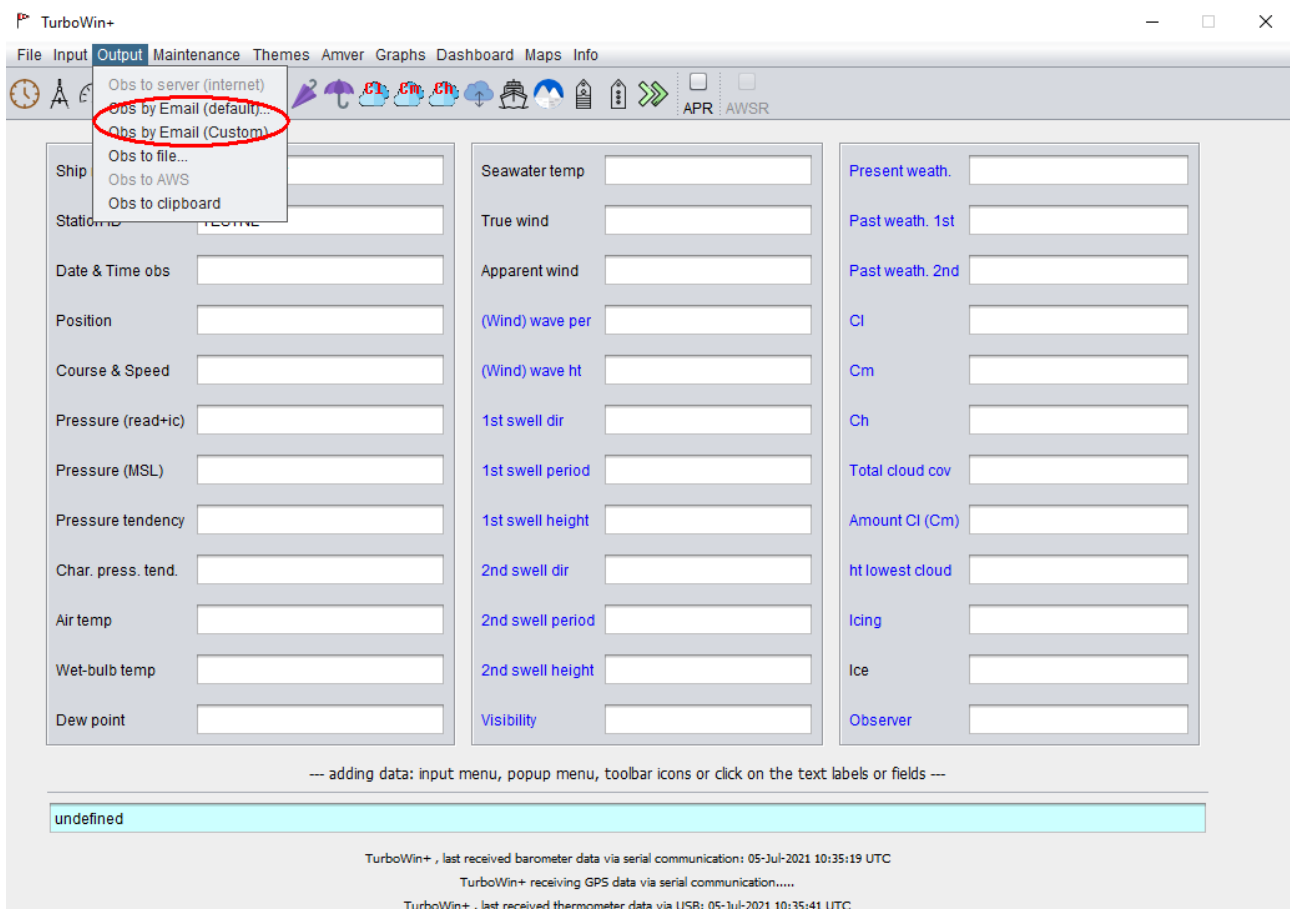


Fig 10 Email output options (surrounded in red); with the email output options available

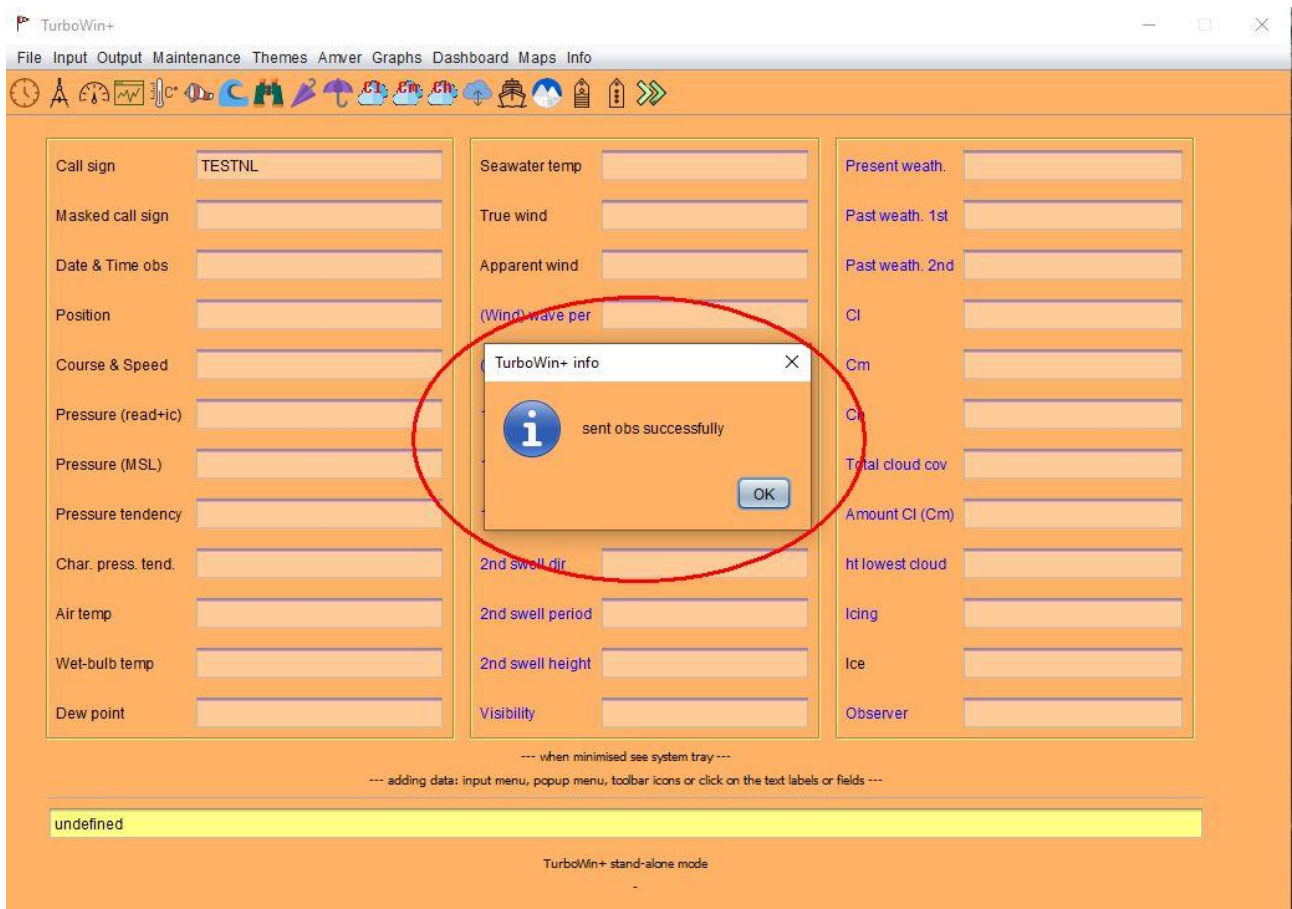


Fig 11 pop-up feedback after sending an observation by Email

```

09-Aug-2019 09:32:30 UTC [GENERAL] created dir C:\NetBeansProjects\turbowin_jwsdist\link\turbowin_jwsbin\logs\turbowin_system
09-Aug-2019 09:32:30 UTC [GENERAL] started TurboWin+ stand-alone mode 3.4.0 beta [64-bit] (build 9-August-2019)
09-Aug-2019 09:32:30 UTC [GENERAL] Java version: OpenJDK 12.0.1
09-Aug-2019 09:32:30 UTC [GENERAL] deleting old (> 3 months) TurboWin+ system logs
09-Aug-2019 09:32:30 UTC [GENERAL] JVM: total designated memory = 2020 MB; current allocated free memory = 106 MB; total allocated memory = 128 MB; used memory = 22 MB; total free memory = 1998 MB
09-Aug-2019 09:39:08 UTC [GENERAL] JVM: total designated memory = 2020 MB; current allocated free memory = 19 MB; total allocated memory = 34 MB; used memory = 15 MB; total free memory = 2005 MB
09-Aug-2019 09:39:09 UTC [GENERAL] stopped TurboWin+ stand-alone mode 3.4.0 beta [64-bit] (build 9-August-2019)
09-Aug-2019 09:39:11 UTC [GENERAL] Java version: OpenJDK 12.0.1
09-Aug-2019 09:39:11 UTC [GENERAL] started TurboWin+ stand-alone mode 3.4.0 beta [64-bit] (build 9-August-2019)
09-Aug-2019 09:39:11 UTC [GENERAL] deleting old (> 3 months) TurboWin+ system logs
09-Aug-2019 09:39:11 UTC [GENERAL] JVM: total designated memory = 2020 MB; current allocated free memory = 106 MB; total allocated memory = 128 MB; used memory = 22 MB; total free memory = 1998 MB
09-Aug-2019 09:39:30 UTC [FORMAT-101] did not find: C:\NetBeansProjects\turbowin_jwsdist\link\turbowin_jwsbin\logs\format_101\teste_hc_TW.exe
09-Aug-2019 09:39:30 UTC [FORMAT-101] start copying complete format_101 module from jar to: C:\NetBeansProjects\turbowin_jwsdist\link\turbowin_jwsbin\logs
09-Aug-2019 09:39:30 UTC [FORMAT-101] success when copying teste_hc_TW.exe from jar to: C:\NetBeansProjects\turbowin_jwsdist\link\turbowin_jwsbin\logs\format_101\teste_hc_TW.exe
09-Aug-2019 09:39:30 UTC [FORMAT-101] success when copying MAWSbin_TW.exe from jar to: C:\NetBeansProjects\turbowin_jwsdist\link\turbowin_jwsbin\logs\format_101\MAWSbin_TW.exe
09-Aug-2019 09:39:30 UTC [FORMAT-101] success when copying S-AWS-101_modl_pilote.csv from jar to: C:\NetBeansProjects\turbowin_jwsdist\link\turbowin_jwsbin\logs\format_101\config\S-AWS-101_modl_pilote.csv
09-Aug-2019 09:39:30 UTC [FORMAT-101] success when copying Data Type.txt from jar to: C:\NetBeansProjects\turbowin_jwsdist\link\turbowin_jwsbin\logs\format_101\config\S-AWS-101_modl_meta.csv
09-Aug-2019 09:39:31 UTC [EMAIL] trying to send obs ( TESTNLYP@C IgAJ'D Xebg x) to martin.stam@home.nl cc martin.stam@knmi.nl from martin.stam@home.nl via SMTP_HOST_SHIP
09-Aug-2019 09:39:31 UTC [EMAIL] did not find: C:\NetBeansProjects\turbowin_jwsdist\link\turbowin_jwsbin\logs\python\email_tbw.exe
09-Aug-2019 09:39:31 UTC [EMAIL] start copying complete Python Email module from jar to: C:\NetBeansProjects\turbowin_jwsdist\link\turbowin_jwsbin\logs
09-Aug-2019 09:39:31 UTC [EMAIL] success when copying python/email_tbw.exe from jar to: C:\NetBeansProjects\turbowin_jwsdist\link\turbowin_jwsbin\logs\python
09-Aug-2019 09:39:34 UTC [EMAIL] mail sent successfully
09-Aug-2019 09:48:02 UTC [GENERAL] JVM: total designated memory = 2020 MB; current allocated free memory = 8 MB; total allocated memory = 30 MB; used memory = 22 MB; total free memory = 1998 MB
09-Aug-2019 09:48:02 UTC [GENERAL] stopped TurboWin+ stand-alone mode 3.4.0 beta [64-bit] (build 9-August-2019)
09-Aug-2019 09:48:05 UTC [GENERAL] Java version: OpenJDK 12.0.1
09-Aug-2019 09:48:05 UTC [GENERAL] started TurboWin+ stand-alone mode 3.4.0 beta [64-bit] (build 9-August-2019)
09-Aug-2019 09:48:05 UTC [GENERAL] deleting old (> 3 months) TurboWin+ system logs
09-Aug-2019 09:48:05 UTC [GENERAL] JVM: total designated memory = 2020 MB; current allocated free memory = 106 MB; total allocated memory = 128 MB; used memory = 22 MB; total free memory = 1998 MB
09-Aug-2019 09:49:10 UTC [FORMAT-101] found: format 101 compression executable (teste_hc_TW.exe)
09-Aug-2019 09:49:12 UTC [EMAIL] trying to send obs ( TESTNLYP@C IgAJ'D Xebg x) to martin.stam@home.nl cc martin.stam@knmi.nl from turbowin.observations@gmail.com via GMAIL_SSL
09-Aug-2019 09:49:12 UTC [EMAIL] found: email_tbw.exe
09-Aug-2019 09:49:16 UTC [EMAIL] mail sent successfully
09-Aug-2019 12:04:29 UTC [GENERAL] started TurboWin+ stand-alone mode 3.4.0.0 beta [64-bit] (build 9-August-2019)
09-Aug-2019 12:04:29 UTC [GENERAL] Java version: OpenJDK 12.0.1
09-Aug-2019 12:04:29 UTC [GENERAL] JVM: total designated memory = 2020 MB; current allocated free memory = 107 MB; total allocated memory = 128 MB; used memory = 21 MB; total free memory = 1999 MB
09-Aug-2019 12:04:29 UTC [GENERAL] deleting old (> 3 months) TurboWin+ system logs
09-Aug-2019 12:04:47 UTC [FORMAT-101] found: format 101 compression executable (teste_hc_TW.exe)
09-Aug-2019 12:04:48 UTC [EMAIL] trying to send obs ( TESTNLYP@C IgAK@D Xebg x) to martin.stam@home.nl cc martin.stam@knmi.nl from turbowin@yahoo.com via Yahoo_TLS
09-Aug-2019 12:04:48 UTC [EMAIL] did not find: C:\NetBeansProjects\turbowin_jwsdist\link\turbowin_jwsbin\logs\python\email_tbw.exe
09-Aug-2019 12:04:48 UTC [EMAIL] start copying complete Python Email module from jar to: C:\NetBeansProjects\turbowin_jwsdist\link\turbowin_jwsbin\logs
09-Aug-2019 12:04:48 UTC [EMAIL] success when copying python/email_tbw.exe from jar to: C:\NetBeansProjects\turbowin_jwsdist\link\turbowin_jwsbin\logs\python
09-Aug-2019 12:04:52 UTC [EMAIL] mail sent successfully

```

fig 12 Extended logging (TurboWin+: Info → System log)

2 Vaisala HMP155 sensor interface

For air temperature, wet-bulb temperature, dew point and relative humidity



Fig 14 Vaisala HMP155

On Windows 7 the required Vaisala HMP155 USB driver (called: “Vaisala USB device”) must be installed manually (from the accompanied Vaisala cd or from the Vaisala web site). It seems that on most Windows 10 pc’s the driver will be installed automatically. Maybe you have to restart the pc after inserting the HMP155 USB cable. On Linux no additional driver required.

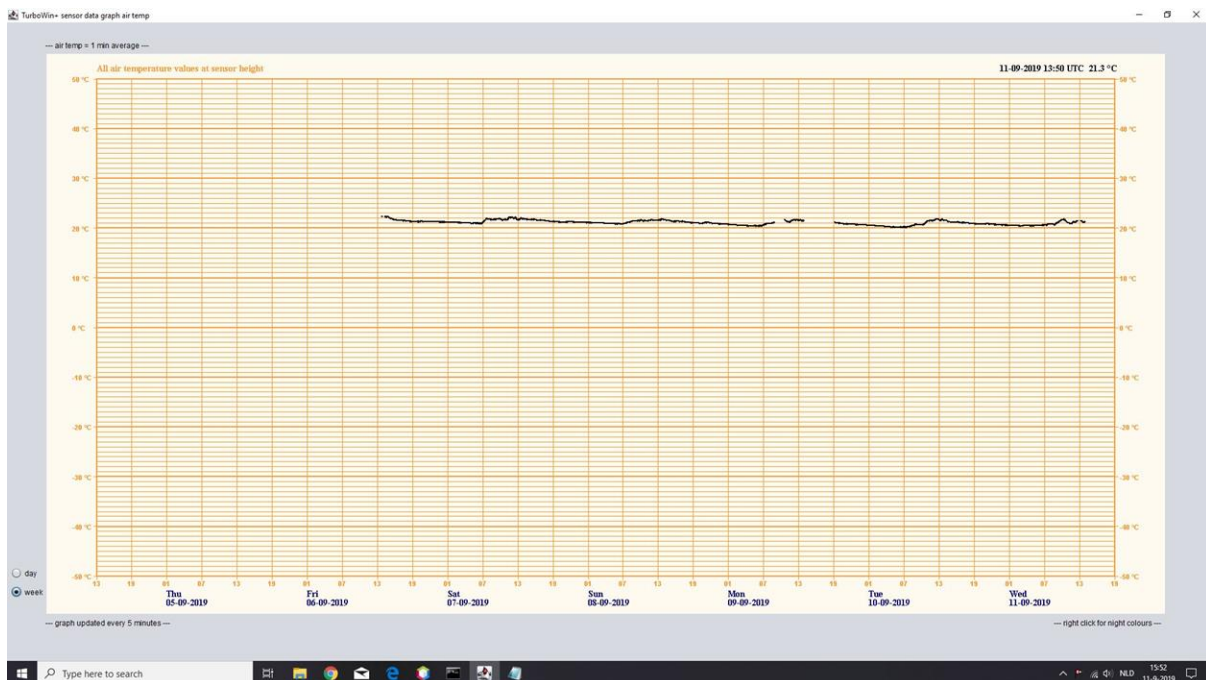


Fig 15 TurboWin+ air temperature graph of measured Vaisala HMP155 data

The HMP155 can be used as one of the required sensors in APTR (Automated Pressure & Temperature reports) mode and manual mode. Observation format #101 or format FM13 via server or email

3 EMOS (Mintaka) interface

-



fig 16 The main components of EMOS: Mintaka Star + StarX + GuardX; For GPS position, air pressure, pressure tendency and characteristic, air temperature, wet-bulb temperature, dew point and relative humidity

EMOS (Enhanced Manual Observation System) can be used in APTR (Automated Pressure & Temperature reports) mode and manual mode. Connection via WiFi or USB. Observation format #101 or format FM13 via server or email.

4 Setting TurboWin+ to APR (or APTR)

4.1 settings

- Select the observation format (Maintenance -> Obs format setting, Fig 17)
- Prepare the output send/upload options (Maintenance -> Email settings or Maintenance → Server settings)
- Select the connected sensors (Maintenance -> Serial/USB/LAN device settings, Fig 19)
- Set the APR options (Maintenance -> APR/APTR/AWSR settings, Fig 20)

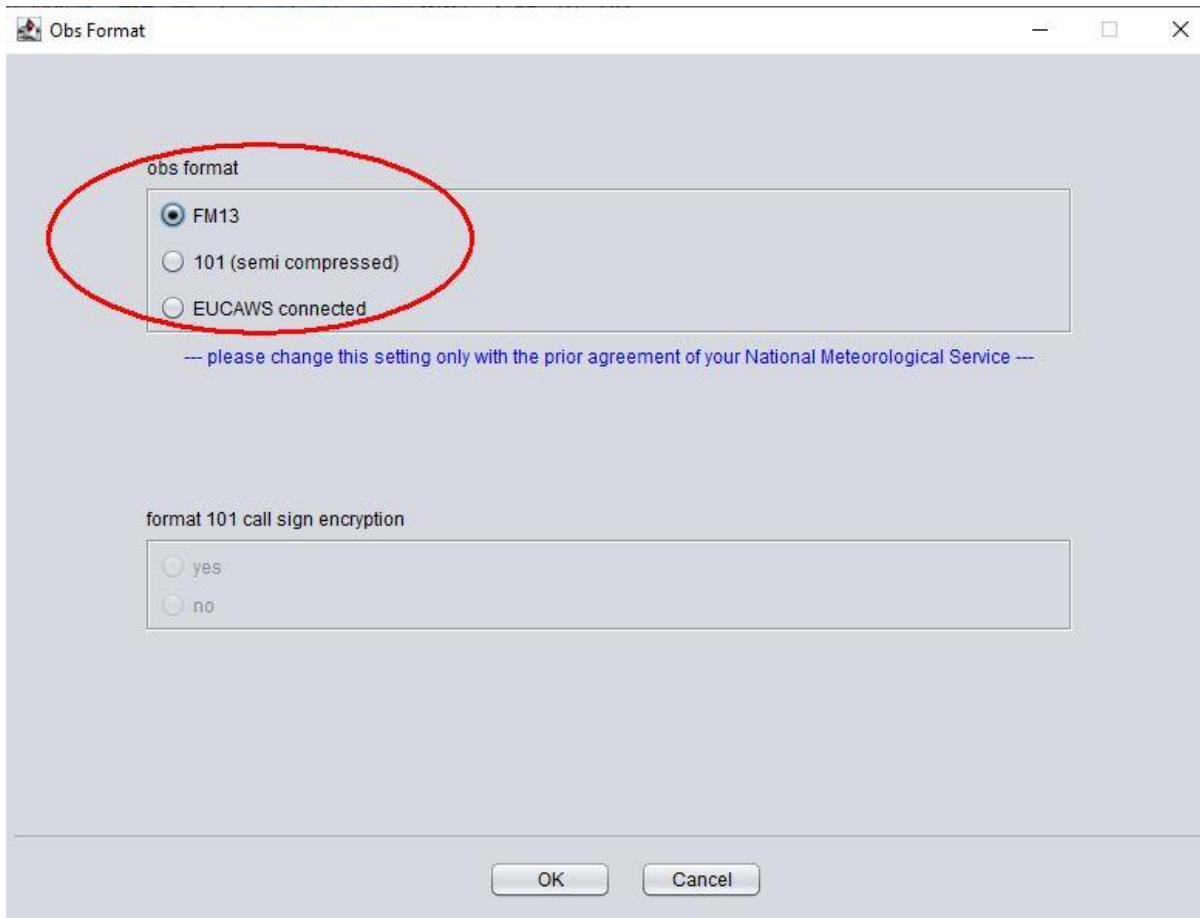


fig 17 selecting obs format

Serial/USB or LAN connection settings

1st meteo instrument

☐ Mintaka Duo USB
☐ OMC-140 AWS LAN
☒ Mintaka Star USB
☐ OMC-140 AWS serial
☐ Mintaka Star WiFi
☐ Vaisala PTB220 serial
☐ Vaisala PTB330 serial
☐ EUCAWS AWS serial
☐ none

2nd meteo instrument

☒ Mintaka StarX WiFi
☐ Vaisala HMP155 USB
☐ none

GPS

☐ GPS (NMEA 0183)
☒ none

1st instrument port settings

bits per second
57600
data bits
8
parity
none
stop bits
1
port number
AUTOMATICALLY

2nd instrument port settings

bps
data bits
parity
stop bits
port number

GPS port settings

bits per second
port number

☐ use RMC sentence (recommended)
☐ use GGA sentence

--- Especially for laptops, set all the energy saving settings, display excepted, to 'never' ---

OK
Cancel

fig 19 selecting connected sensors

AP[&T]R and AWSR settings

AP[&T]R (Automated Pressure [& Temperature] Reports)

☒ report pressure [& temp] automatically (barometer* + GPS)

☒ 1 hour ☐ 3 hours ☐ 6 hours

* recommended: static head or flexible tube or WiFi sensor to measure outside pressure

AWSR (Automatic Weather Station Reports)

☐ report all measured data automatically (AWS connected)

☐ 1 hour ☐ 3 hours ☐ 6 hours

AP[&T]R / AWSR send method

☐ Server Met Center* ☒ Custom email**

*Maintenance -> Server settings **Maintenance -> Email settings

AP[&T]R

normal steaming draft (metres, e.g. 12.6) [0.0 - 50.0]

barometer instrument correction** (hPa, e.g. -0.1) [-4.0 - 4.0] ** average over the whole calibration range (this is NOT the height corr)

--- minimise TurboWin+ after AP[&T]R or AWSR settings are set ---

--- enable TurboWin+ autostart to continue automatic reporting after a computer restart ---

OK Cancel

fig 20 selecting the specific APR (APTR) options

4.2 Adding manual observed data when in APR/APTR mode

This is very easy. Open the appropriate meteorological parameter input screen, add the data and click OK (Fig 21). No further actions required. The inserted data will be automatically added to the observation with the next scheduled upload.

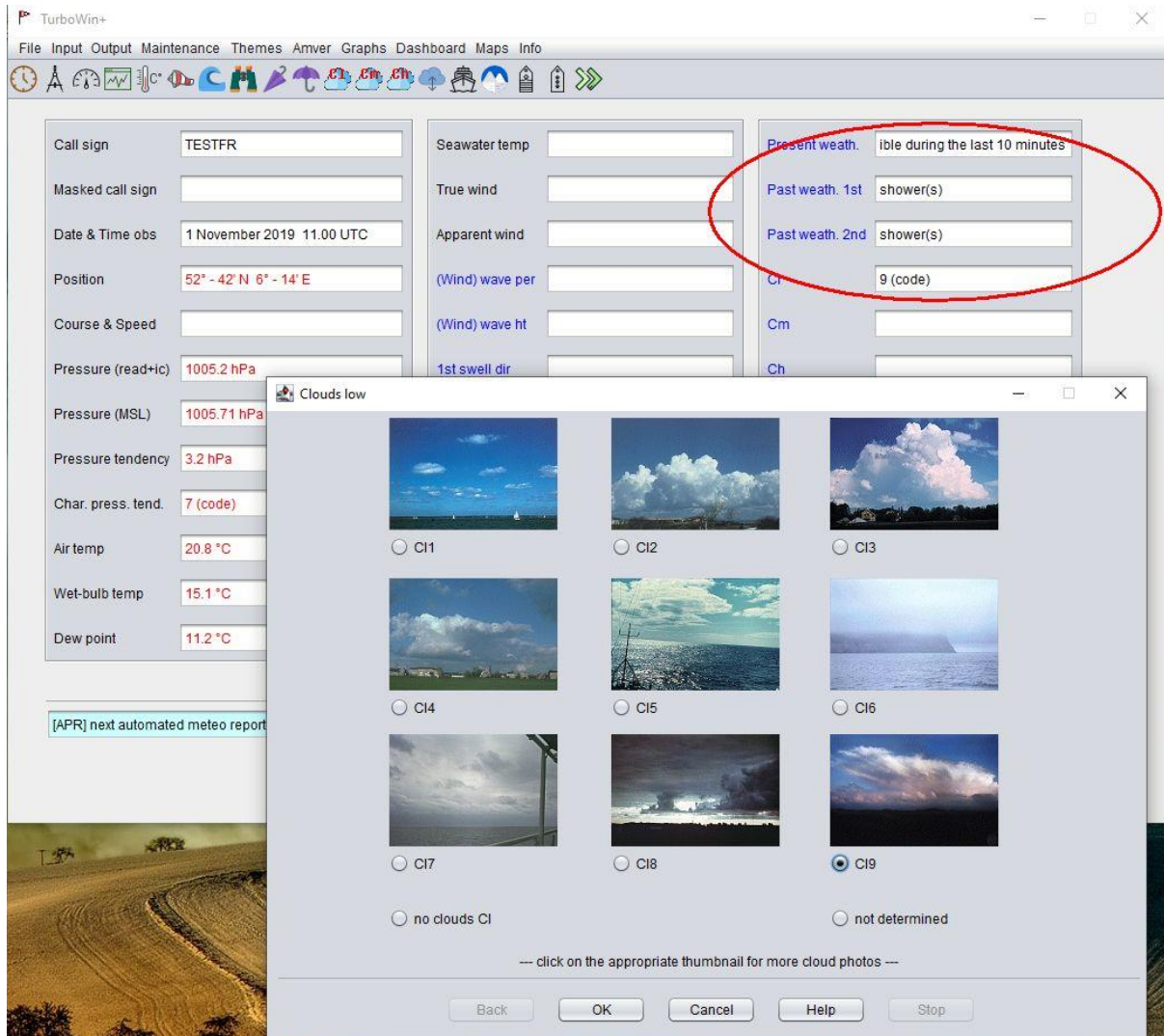


fig 21 adding manual observed data in APR/APTR mode

4.3 Switching between automatic modes APR/APTR and semi automatic mode

When in APR/APTR mode uncheck the AP[&T]R thickbox “report pressure [& temp] auto.” and click OK (Fig 22). Not necessary to restart the computer. Now all output options and input screens are available as in the full manual mode or semi automatic mode. Check the thickbox again to continue with APR/APTR. Note, you also can switch APR on and off with the buttons on the main screen (see paragraph 4.4)

AP[&T]R and AWSR settings

AP[&T]R (Automated Pressure [& Temperature] Reports)

☒ report pressure [& temp] automatically (barometer* + GPS)

☒ 1 hour ☐ 3 hours ☐ 6 hours

* recommended: static head or flexible tube or WiFi sensor to measure outside pressure

AWSR (Automatic Weather Station Reports)

☐ report all measured data automatically (AWS connected)

☐ 1 hour ☐ 3 hours ☐ 6 hours

AP[&T]R / AWSR send method

☐ Server Met Center* ☒ Custom email**

* Maintenance -> Server settings **Maintenance -> Email settings

AP[&T]R

normal steaming draft (metres, e.g. 12.6) [0.0 - 50.0]

barometer instrument correction** (hPa, e.g. -0.1) [-4.0 - 4.0] ** average over the whole calibration range (this is NOT the height corr)

--- minimise TurboWin+ after AP[&T]R or AWSR settings are set ---

--- enable TurboWin+ autostart to continue automatic reporting after a computer restart ---

OK Cancel

fig 22 switching between APR (APTR) and manual mode on settings screen

4.4 AP[&T]R and AWSR buttons on main screen

If the necessary setting were once made before (Maintenance → APR/APTR/AWSR settings). You also can turn on and off the AP[&T]R mode with one mouse click (Fig 23). Same for the AWSR mode.

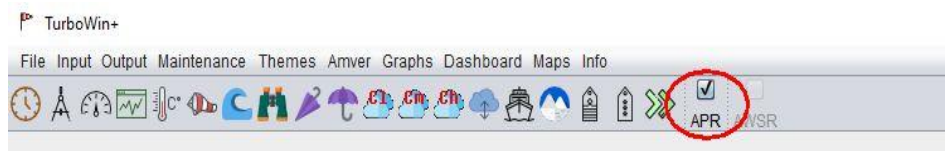


fig 23 switching between APR (APTR) and manual mode on main screen

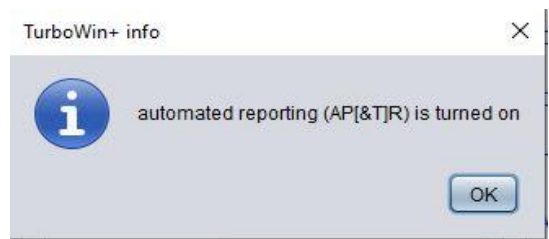


fig 24 APR (APTR) turned on message

5 Station ID

It was decided that there will be unique identifiers for marine platforms and instrument systems / packages on ships

See Fig 28 for the TurboWin+ station data input screen with the specific station ID input filed in red

Station data

ship data

ship name: Happy tester

IMO number: 1234567

station ID*: TESTNL

* consult your PMO or National Meteorological Service

recruiting country

MOZAMBIQUE MZ
MYANMAR MM
NAMIBIA NA
NAURU NR
NEPAL NP
NETHERLANDS NL
NETHERLANDS ANTILLES AN
NEW CALEDONIA NC

temperatures meta data

air temp exposure: sling psychrom. (selected), marine screen

sea water temp exposure: intake (selected), bucket, hull contact sens., trailing thermistor, "through hull" sensor, radiation therm., bait tanks therm., other

wind meta data

estimated; true speed and true direction (selected)
measured; apparent speed + app. dir. (OFF THE BOW, clockwise)*
measured; true speed and true direction

max. height deck cargo above SLL** (metres, [0 - 100], rounded): 20

difference SLL and WL (water line) (metres, [-10 - 50]****, rounded): 2

wind speed units source (estimated/measured): knots (selected), m/s*

wind speed units display (graphs/dashboards): knots (selected), m/s

average height anemometer above WL (metres, [0-100], rounded)***: 12

* inter alia when an AWS is connected ** Summer Load Line
*** only when an AWS is connected **** negative if SLL is below WL

air pressure meta data

height of the barometer above SLL (metres, e.g. 20.8): 20.8

distance of bottom of the keel to SLL (metres, e.g. 9.1): 9.1

does the reading indicate MSL pressure*: yes, no* (selected)

* always "no" when a barometer or an AWS is connected

OK Cancel import

fig 28 Station data input screen

6 Dashboards

Dashboards available in AWS (Automatic Weather Station) connected mode in APR/APTR (Automated Pressure [&Temperature] Reports) mode

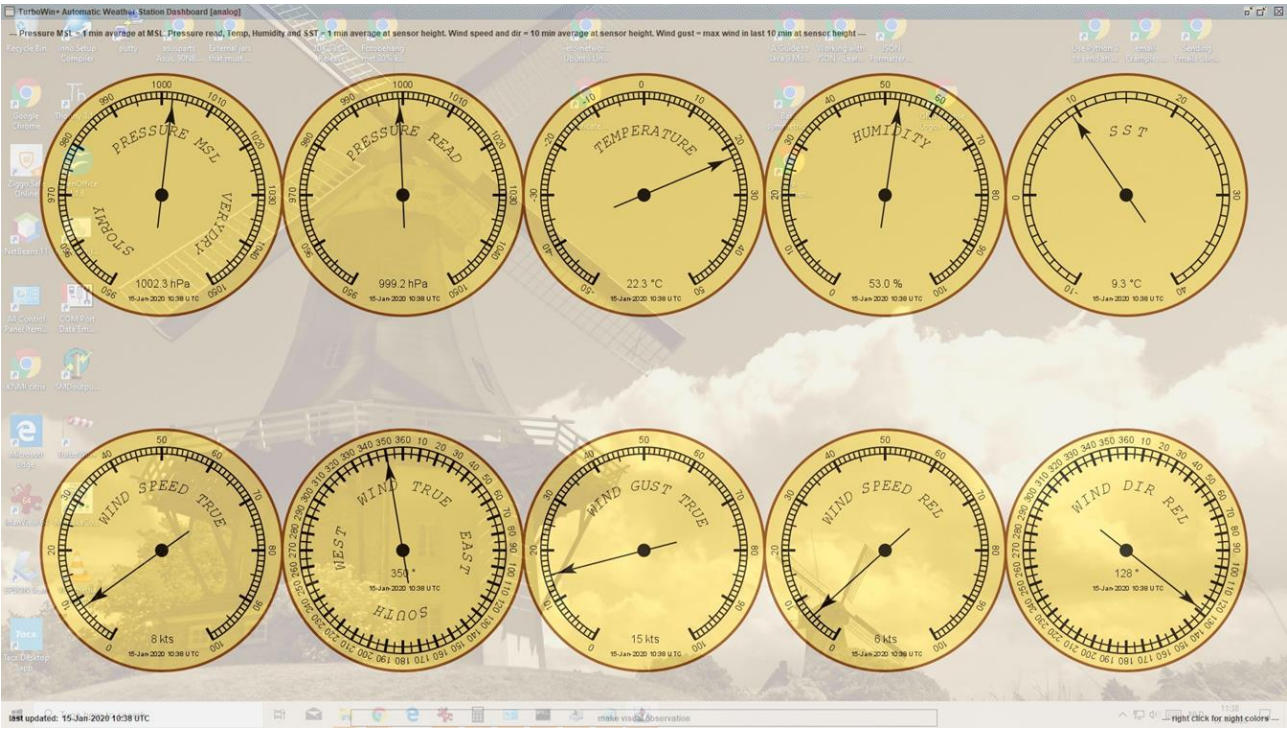


fig 30 AWS analog dashboard (in transparent screen mode)

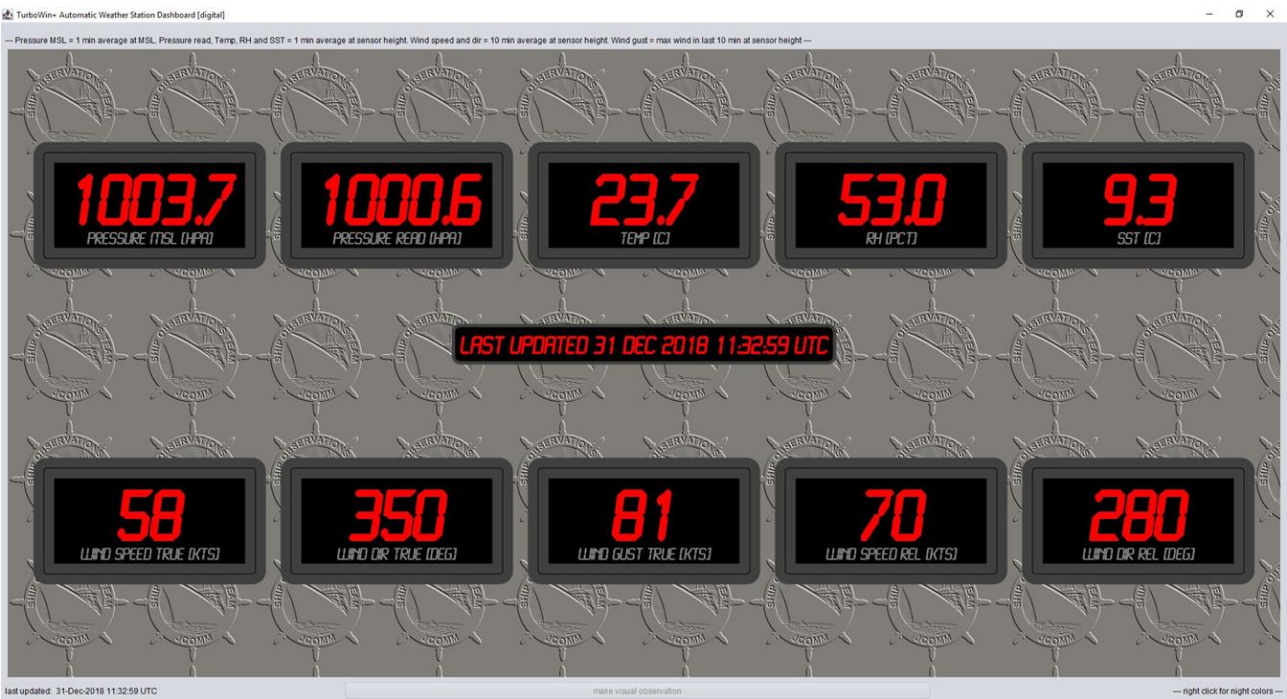


fig 31 AWS digital dashboard

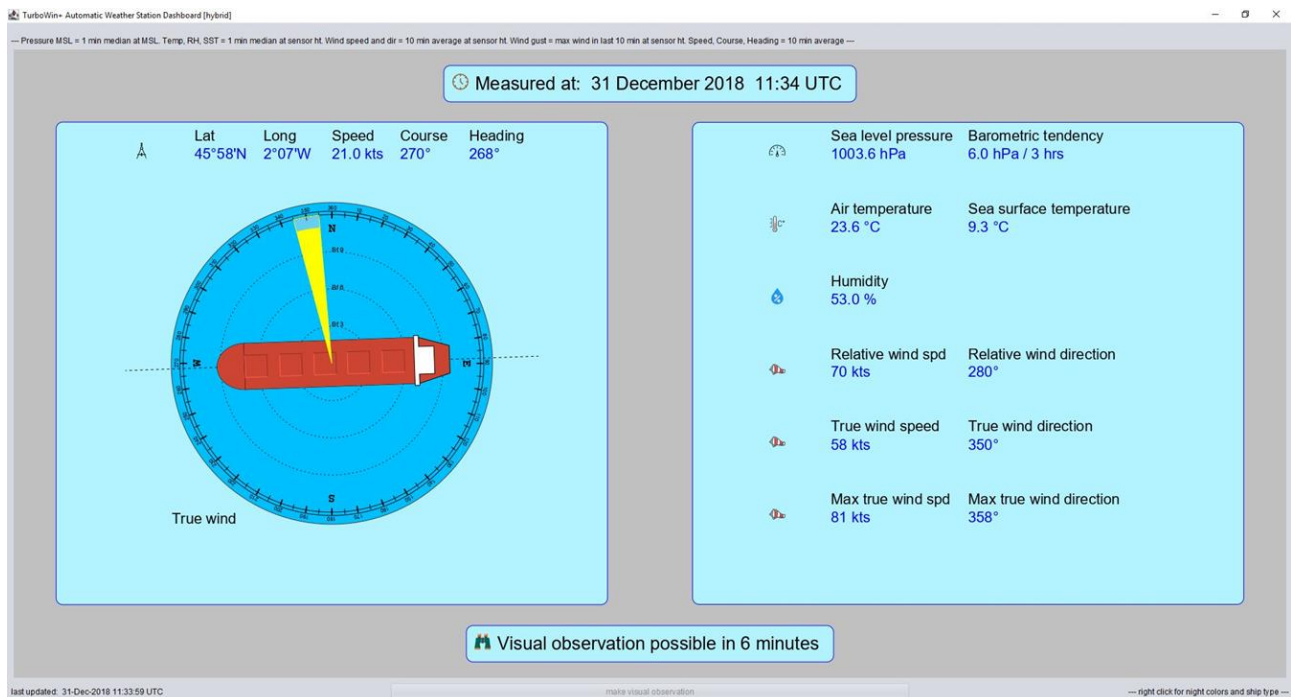


fig 32 AWS hybrid dashboard

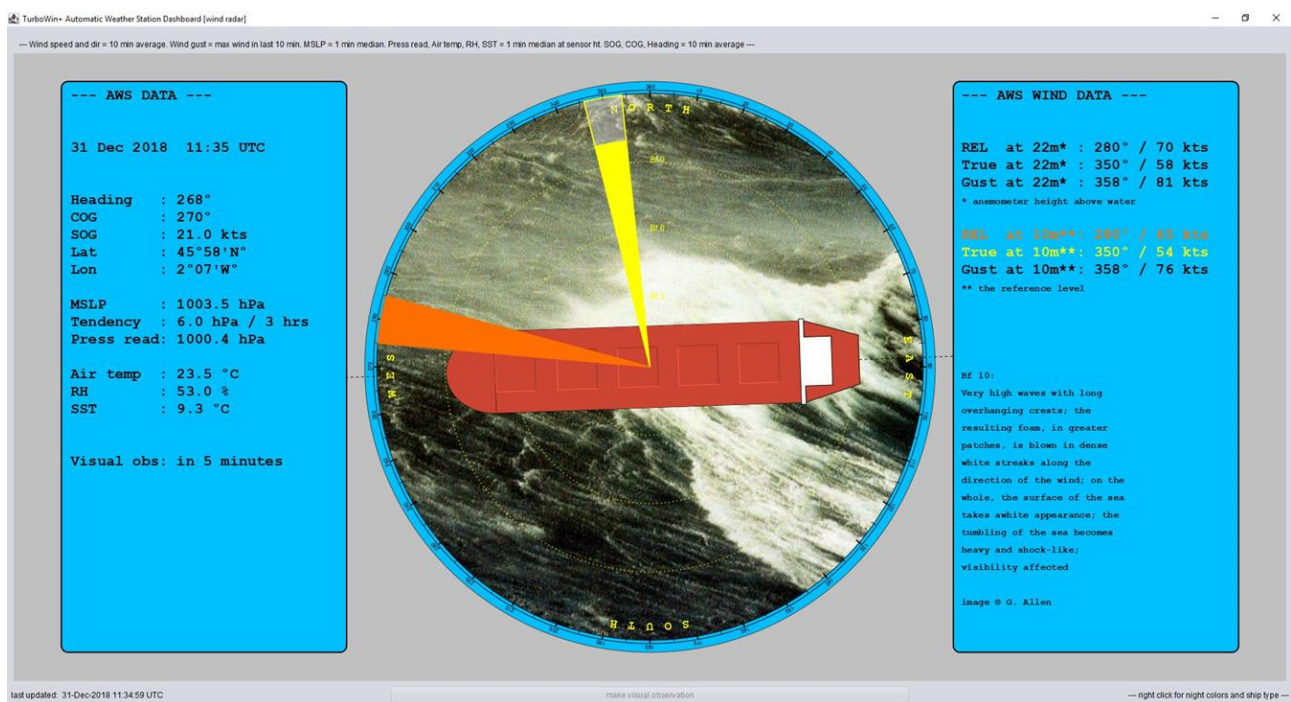


fig 33 AWS wind radar dashboard



fig 34 APR meteo radar dashboard (day colours)

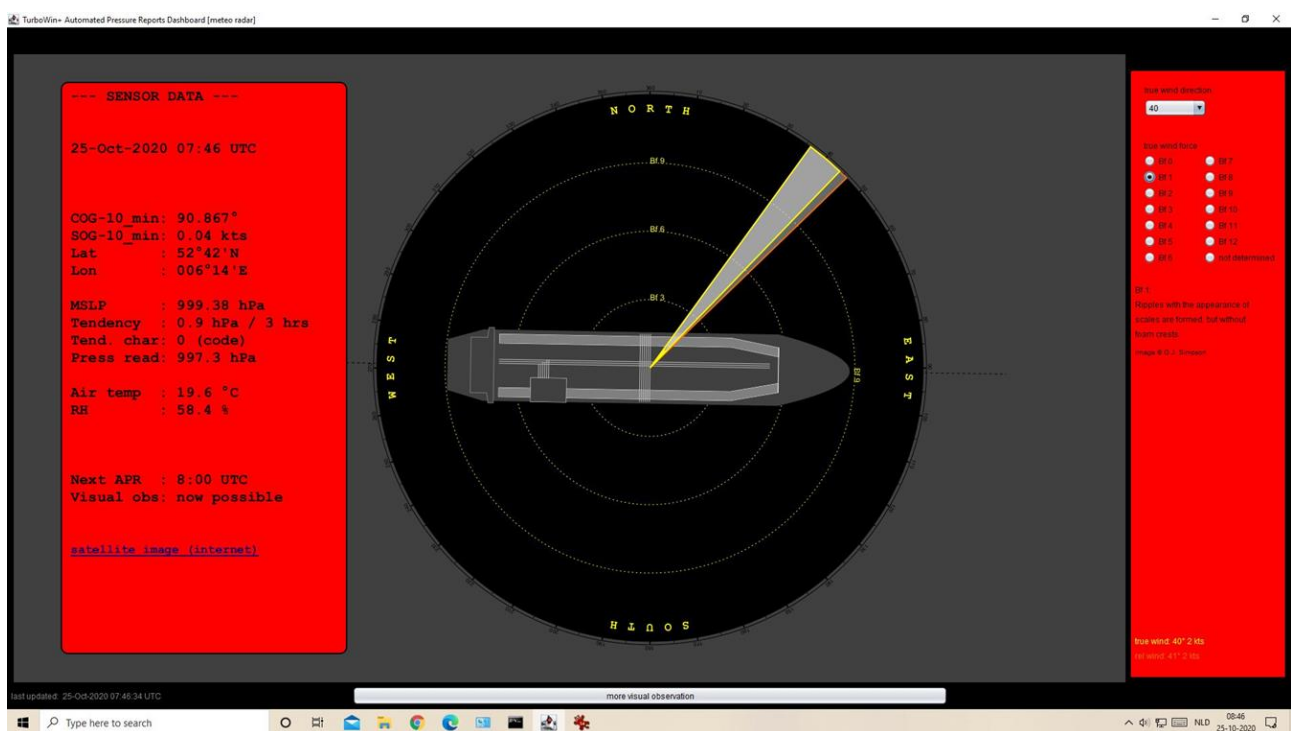


fig 35 APR meteo dashboard (night colours)

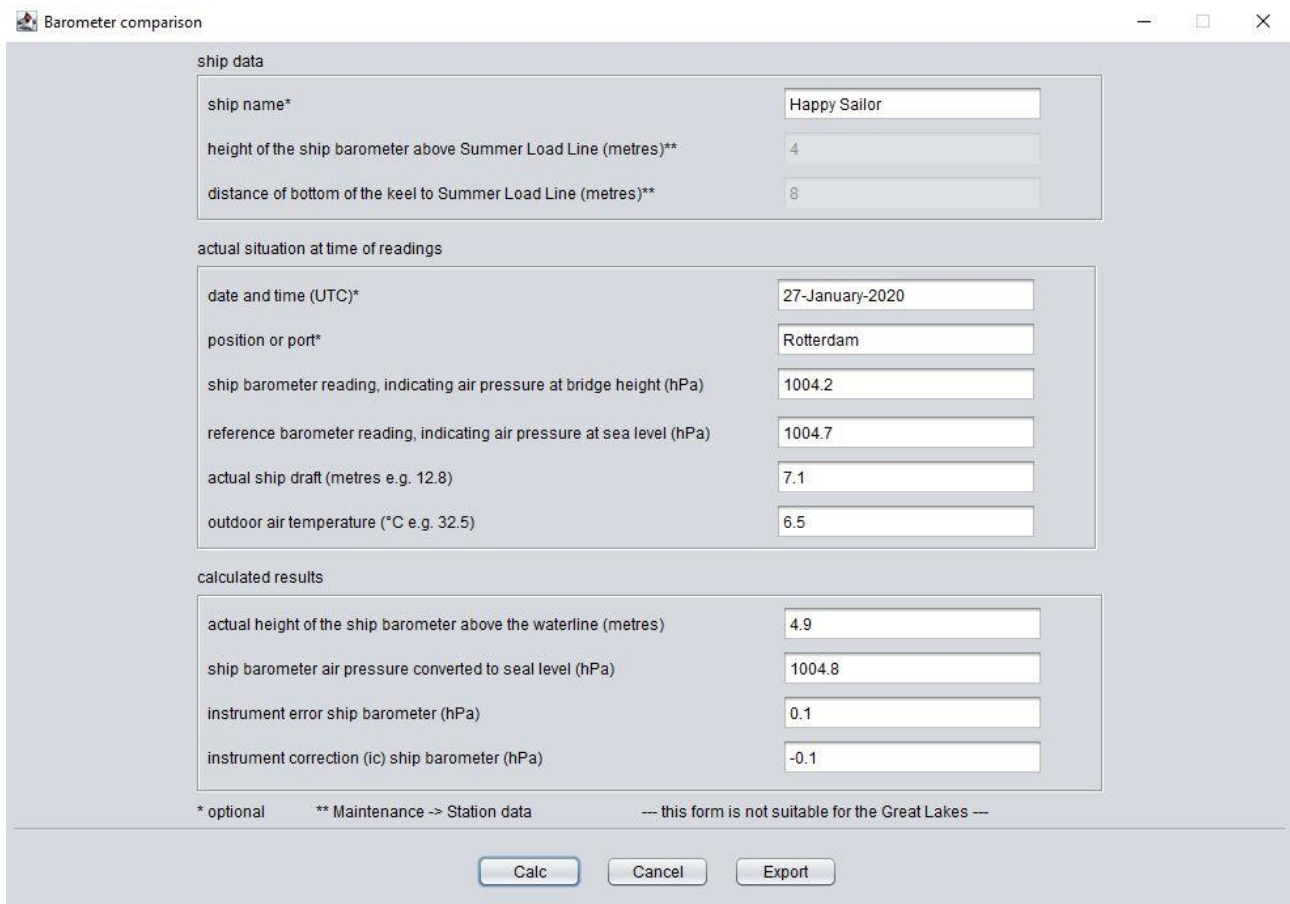
Selectable ship types in dashboards:



fig 36 dashboard ship types

7 Barometer comparison form (barometer self-check form)

A barometer comparison form was added. Select: Info → barometer comparison



The screenshot shows a software window titled "Barometer comparison" with standard window controls (minimize, maximize, close) in the top right corner. The form is organized into three main sections: "ship data", "actual situation at time of readings", and "calculated results".

ship data

ship name*	Happy Sailor
height of the ship barometer above Summer Load Line (metres)**	4
distance of bottom of the keel to Summer Load Line (metres)**	8

actual situation at time of readings

date and time (UTC)*	27-January-2020
position or port*	Rotterdam
ship barometer reading, indicating air pressure at bridge height (hPa)	1004.2
reference barometer reading, indicating air pressure at sea level (hPa)	1004.7
actual ship draft (metres e.g. 12.8)	7.1
outdoor air temperature (°C e.g. 32.5)	6.5

calculated results

actual height of the ship barometer above the waterline (metres)	4.9
ship barometer air pressure converted to seal level (hPa)	1004.8
instrument error ship barometer (hPa)	0.1
instrument correction (ic) ship barometer (hPa)	-0.1

At the bottom of the form, there are three buttons: "Calc", "Cancel", and "Export".

* optional ** Maintenance -> Station data --- this form is not suitable for the Great Lakes ---

fig 41 barometer comparison form

8 Open source

TurboWin+ is free software and open-source software (<https://www.gnu.org/licenses/gpl-3.0.en.html>)

This allow users to run the software for any purpose as well as to study, change, and distribute it and any adapted versions. Anyone is freely licensed to use, copy, study, and change the software in any way, and the source code is openly shared on GitLab. (GPLv3 open-source license)

To emphasis the open source nature of TurboWin+ the official GNUv3 logo was added to the Info → About box.

Permissions:

- commercial use
- modification
- distribution
- patent use and private use

limitations:

- liability
- warranty

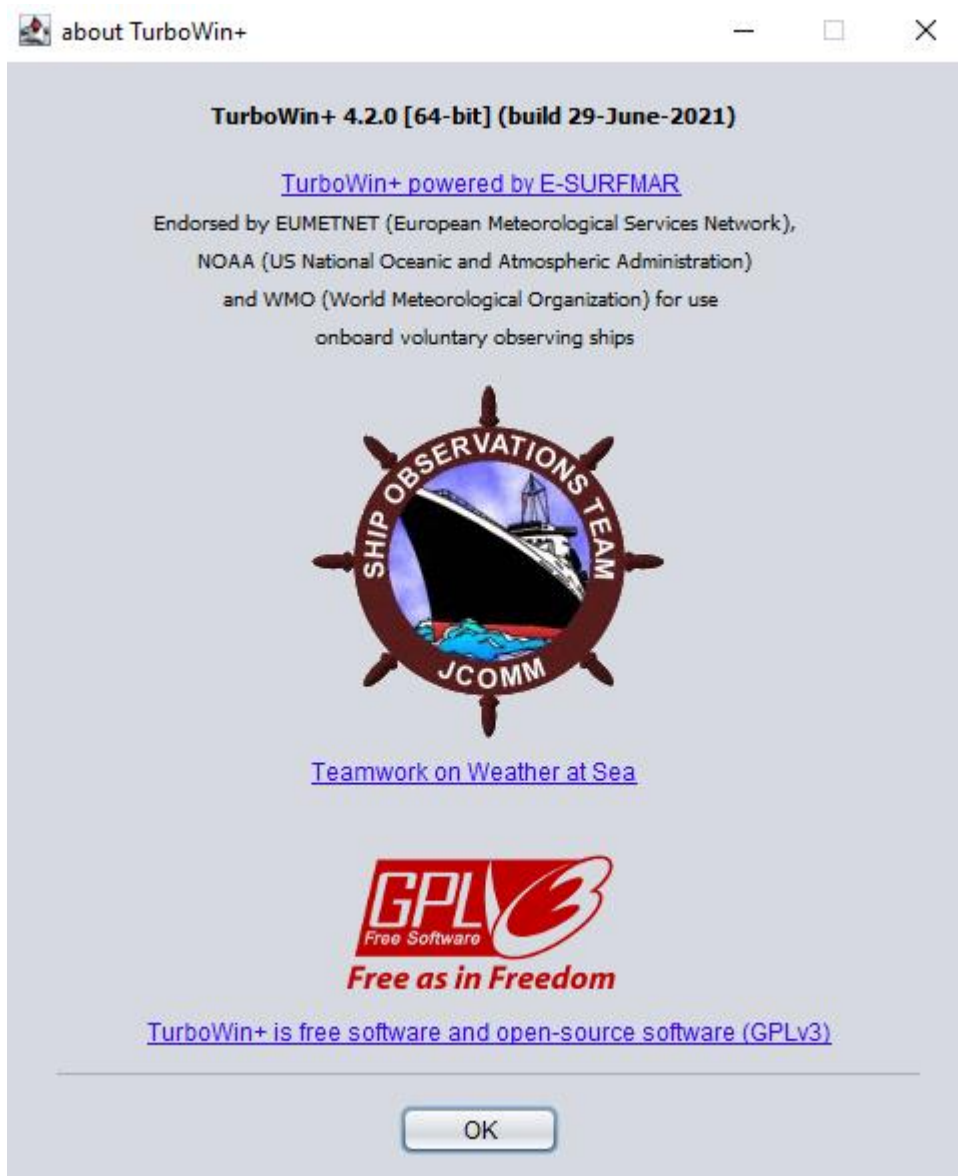


fig 42 About TurboWin+

9 Expert Tips

- 9.1] Memory check
- 9.2] COM ports (Windows)
- 9.3] USB hub
- 9.4] Mintaka Star bridged connection
- 9.5] Verbose mode
- 9.6] Check more than one instance of TurboWin+ is running
- 9.7] Email log
- 9.8] Format #101 logging
- 9.9] Linux (64bit) installation
- 9.10] Obs to server (Rest API)
- 9.11] SRM (Serial communication Recovery Mode)
- 9.12] Check your observations
- 9.13] Autostart

9.1) Memory check

At start up the TurboWin+ memory load of the pc was written to the TurboWin+ system log (Info → System log). See Fig 43

```
05-Feb-2020 12:13:49 UTC [GENERAL] started TurboWin+ 4.0.c beta [64-bit] (build 5-February-2020)
05-Feb-2020 12:13:49 UTC [GENERAL] Java version: OpenJDK-13.0.1
05-Feb-2020 12:13:49 UTC [GENERAL] JVM: total designated memory = 2020 MB; current allocated free memory = 114 MB; total allocated memory = 128 MB; used memory = 14 MB; total free memory = 2006 MB
05-Feb-2020 12:13:49 UTC [GENERAL] deleting old (< 3 months) TurboWin+ system logs
05-Feb-2020 12:13:49 UTC [GENERAL] deleting old sensor data files
05-Feb-2020 12:13:49 UTC [WIFI] start listening
```

fig 43 example memory load TurboWin+ at start-up

You can also check the memory load when TurboWin+ is running (snapshot) by clicking on one of the application exit options but not confirming the exit (click on the NO button), see fig 44



fig 44 TurboWin+ Exit confirmation message

After clicking NO, select Info -> System log. A memory load line was again written in the log, see fig 45

```
10-Feb-2020 10:00:09 UTC [APR] GPS error (no GPS info in last records or no sensor data file available / record formatting error / last saved record obsolete / checksum not ok)
10-Feb-2020 10:23:50 UTC [GENERAL] JVM: total designated memory = 2020 MB; current allocated free memory = 24 MB; total allocated memory = 48 MB; used memory = 24 MB; total free memory = 1996 MB
10-Feb-2020 10:29:11 UTC [GENERAL] JVM: total designated memory = 2020 MB; current allocated free memory = 27 MB; total allocated memory = 48 MB; used memory = 21 MB; total free memory = 1999 MB
```

fig 45 example snapshot memory load TurboWin+

9.2) COM ports (Windows)

“In case you are facing with trouble with your COM ports in use that stacking, these brief steps below will help you to clear them out (for example, when you found myself needing to connect via a USB to Serial adapter to a server and noticed that 35 COM ports showed in use). Somehow the COM ports were not getting cleared out.

The fact is that when you connect a new COM or a USB device (such as smartphone, modem, RS232 to USB converters, etc.), Windows detects newly connected device using Plug-and-Play (PnP) and creates a new virtual COM port with a number from 1 to 255. When you remove this device, the created COM port will not be deleted automatically. If you reconnect this device to a computer, the system re-assigned to it the reserved COM port number. But for any new COM device is assigned the first unallocated (free) COM port number. Thus, over time, a large number of reserved COM ports (in “in use” state) for certain COM or USB devices may appear in the system, although the devices themselves may not be connected to your computer. These ports by-default are not visible to the user in Device Manager”. Continue reading : <https://theitbros.com/how-to-delete-com-ports-in-use/>

9.3) USB hub

If you use a USB hub it is highly recommended to use a powered hub to avoid potential communication errors. Powered hubs (also called active hubs) come with their own AC adapter and plug into the mains (Fig 46)



fig 46 powered USB hub

9.4) Mintaka Star Bridged connection

If the pc onboard is connected with a Ethernet cable to the internet you can connect the Mintaka Star with an USB cable to the pc. But, as an alternative, you can also use a network bridge by setting the Mintaka Star to WiFi mode (Access Point mode, Star creates its own network) and creating a network bridge under Windows in the Network Connections section (Fig 47)

Note

After every pc restart you have to reconnect the bridge!

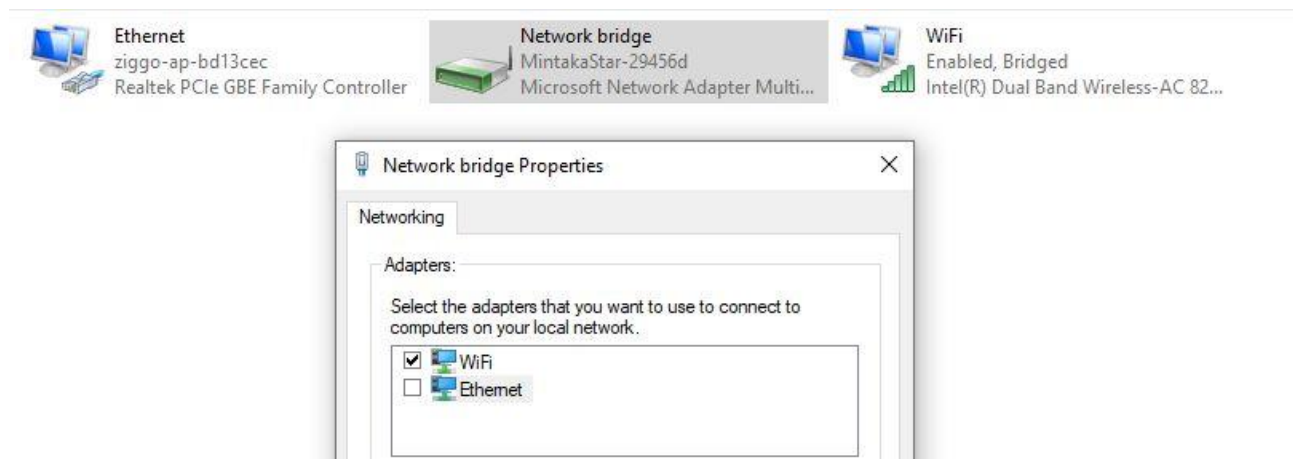


Fig 47 Windows control panel network connections; Mintaka Star via a network bridge

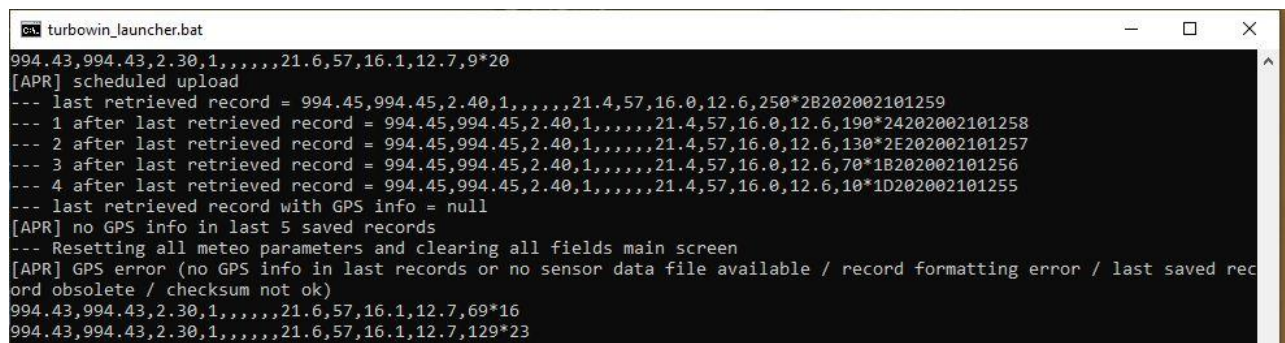
9.5) Verbose mode

For test purposes it is possible to start TurboWin+ in verbose mode by means of invoking “turbowin_launcher.bat”

(e.g. Full path C:\Program Files (x86)\TurboWin+\bin\turbowin_launcher.bat). A console window will be opened (Fig 48).

note

Do not use this option in operational mode



```
turbowin_launcher.bat
994.43,994.43,2.30,1,,,,,21.6,57,16.1,12.7,9*20
[APR] scheduled upload
--- last retrieved record = 994.45,994.45,2.40,1,,,,,21.4,57,16.0,12.6,250*2B202002101259
--- 1 after last retrieved record = 994.45,994.45,2.40,1,,,,,21.4,57,16.0,12.6,190*24202002101258
--- 2 after last retrieved record = 994.45,994.45,2.40,1,,,,,21.4,57,16.0,12.6,130*2E202002101257
--- 3 after last retrieved record = 994.45,994.45,2.40,1,,,,,21.4,57,16.0,12.6,70*1B202002101256
--- 4 after last retrieved record = 994.45,994.45,2.40,1,,,,,21.4,57,16.0,12.6,10*1D202002101255
--- last retrieved record with GPS info = null
[APR] no GPS info in last 5 saved records
--- Resetting all meteo parameters and clearing all fields main screen
[APR] GPS error (no GPS info in last records or no sensor data file available / record formatting error / last saved record obsolete / checksum not ok)
994.43,994.43,2.30,1,,,,,21.6,57,16.1,12.7,69*16
994.43,994.43,2.30,1,,,,,21.6,57,16.1,12.7,129*23
```

fig 48 TurboWin+ console window

9.6) Check more than one instance of TurboWin+ (Windows OS)

TurboWin+ is based on Java. It is not easy for Java programs to check that not more than one instance of an application is running. A well-known and generally accepted approach in the Java community is to claim a never used network computer port (e.g. 12345). So a second instance of TurboWin+ is trying to do the same but get a message that the network port is already occupied (Fig 49)

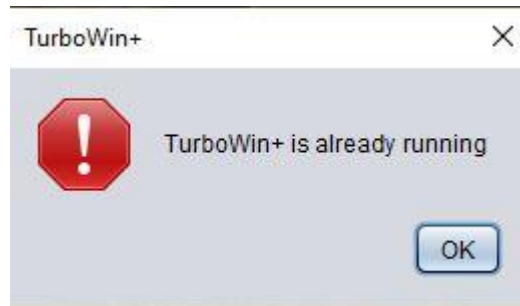


fig 49 TurboWin+ was invoked but was already running

Now is it theoretically possible that another Java program is doing the same, or port 12345 is closed by security policy. In the latter case please add port 12345 to the firewall exception list. In verbose mode, for testing, it is possible to start TurboWin+ with another random (multiple-instances-check) network port. This can be done by creating a shortcut to "turbowin_launcher.bat" and adding the new network port number on the target line (Fig 50).

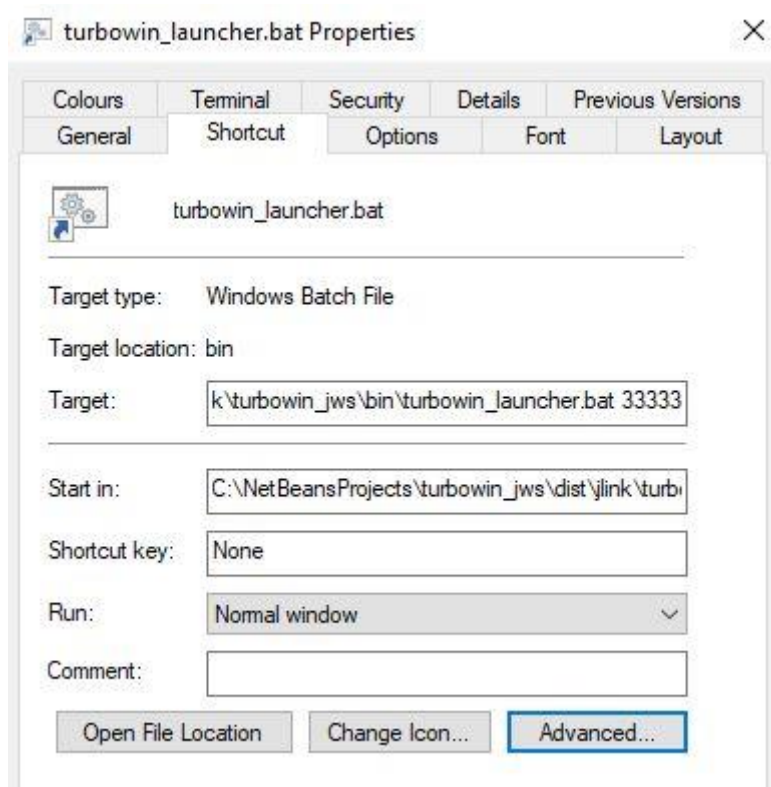


fig 50 TurboWin+ launcher shortcut with network port 33333 as alternative check multiple instances running

```

Ca. turbowin_launcher.bat
--- Argument 0: 33333
user dir = C:\NetBeansProjects\turbowin_jws\dist\jlink\turbowin_jws\bin
--- Resetting all meteo parameters and clearing all fields main screen
-- server port for checking multiple instances running = 33333
[GENERAL] started turbowin+ 4.0.0 beta [64-bit] (build 5-February-2020)
[GENERAL] Java version: OpenJDK 13.0.1

```

fig 51 TurboWin+ console window, network port 33333 used for checking more than one TurboWin+ instance is running

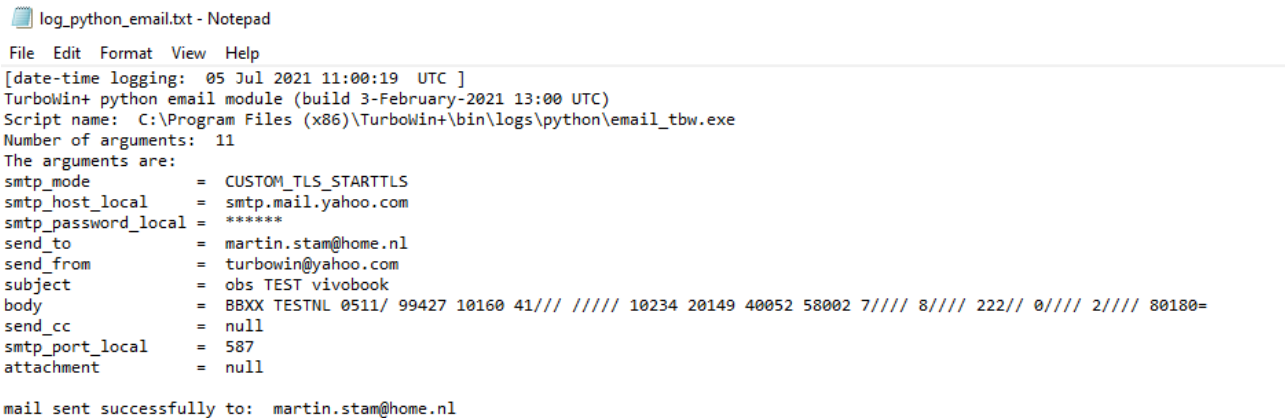
The console window will show the result, see Fig 51.

note

Do not use this option, invoking TurboWin+ via “turbowin_launcher.bat”, in operational mode

9.7) Email log

Besides the default logging in the TurboWin+ system log (Info → System log) you can also check the dedicated email log if the mail was sent with the Custom email option (Maintenance → Email settings). Location of this log: ../logs/python/log_python_email.txt. See Fig 52



```
log_python_email.txt - Notepad
File Edit Format View Help
[date-time logging: 05 Jul 2021 11:00:19 UTC ]
TurboWin+ python email module (build 3-February-2021 13:00 UTC)
Script name: C:\Program Files (x86)\TurboWin+\bin\logs\python\email_tbw.exe
Number of arguments: 11
The arguments are:
smtp_mode           = CUSTOM_TLS_STARTTLS
smtp_host_local     = smtp.mail.yahoo.com
smtp_password_local = *****
send_to             = martin.stam@home.nl
send_from           = turbowin@yahoo.com
subject             = obs TEST vivobook
body                = BBXX TESTNL 0511/ 99427 10160 41/// // 10234 20149 40052 58002 7/// 8/// 222// 0/// 2/// 80180=
send_cc             = null
smtp_port_local     = 587
attachment          = null

mail sent successfully to: martin.stam@home.nl
```

fig 52 dedicated email log

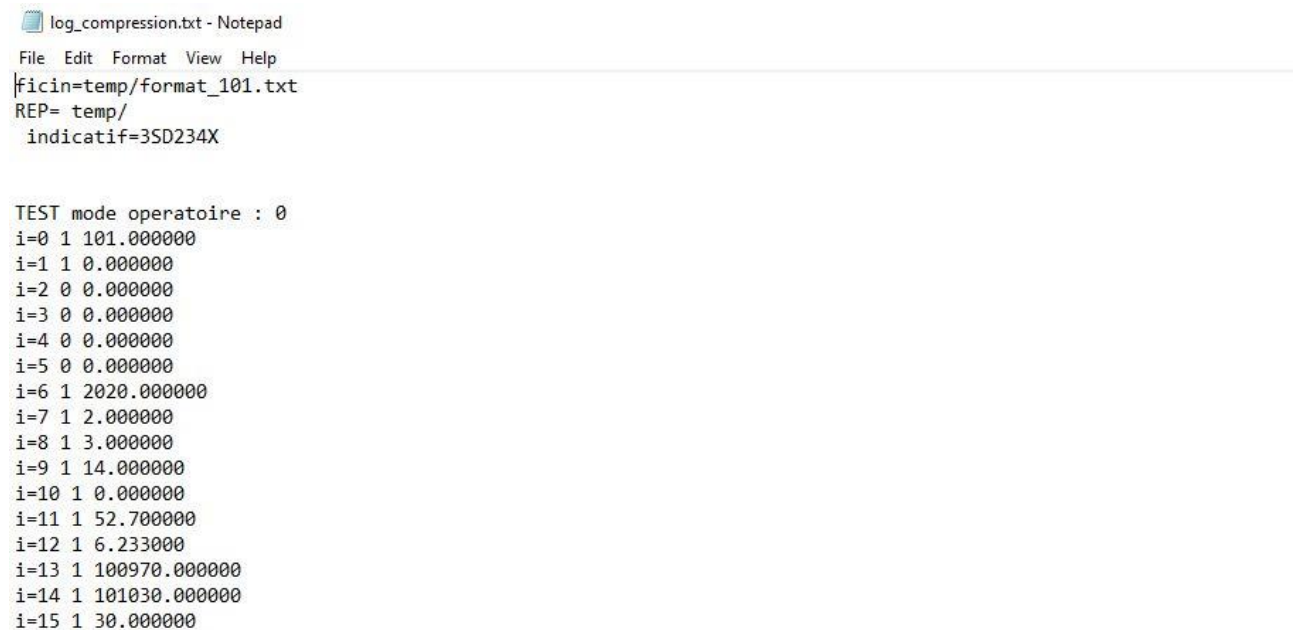
9.8) Format #101 logging

There are dedicated format#101 generated logs (although partly in the French language, they could be helpful when trouble shooting) available after one of the output options in format#101 mode was invoked. These logs are always generated in format #101 mode.

Locations:

../logs/format_101/log/log_compression.txt (Fig 43)

../logs/format_101/log/log_decompression.txt (Fig 44)



```
log_compression.txt - Notepad
File Edit Format View Help
|ficin=temp/format_101.txt
REP= temp/
  indicatif=3SD234X

TEST mode operatoire : 0
i=0 1 101.000000
i=1 1 0.000000
i=2 0 0.000000
i=3 0 0.000000
i=4 0 0.000000
i=5 0 0.000000
i=6 1 2020.000000
i=7 1 2.000000
i=8 1 3.000000
i=9 1 14.000000
i=10 1 0.000000
i=11 1 52.700000
i=12 1 6.233000
i=13 1 100970.000000
i=14 1 101030.000000
i=15 1 30.000000
```

fig 53 -part of- format#101 compression log


```
log_decompression.txt - Notepad
File Edit Format View Help
005002 - - 52.700000 - Latitude (coarse accuracy)
006002 - - 6.230000 - Longitude (coarse accuracy)
010004 - - 100970.000000 - Pressure at barometer height
010051 - - 101030.000000 - Pressure reduced to mean sea level
010061 - - 30.000000 - 3-hour pressure change
010063 - - 0.000000 - Characteristic of pressure tendency
011001 - - ---ABSENT--- - True wind direction clockwise from the north
011002 - - ---ABSENT--- - True wind speed
011099 - - ---ABSENT--- - Relative wind direction clockwise from the bow
011100 - - ---ABSENT--- - Relative wind speed
011041 - - ---ABSENT--- - Maximum wind gust speed
011043 - - ---ABSENT--- - Maximum wind gust direction
012101 - - 294.250000 - Air temperature
012102 - - 289.050000 - Wet bulb temperature
012103 - - 285.650000 - Dew point temperature
013003 - - 58.400000 - Relative humidity
022042 - - ---ABSENT--- - Sea temperature
020001 - code - ---ABSENT--- - Horizontal visibility
020003 - - ---ABSENT--- - Present weather
020004 - - ---ABSENT--- - Past weather 1
020005 - - ---ABSENT--- - Past weather 2
020011 - N - ---ABSENT--- - Total cloud cover
020011 - Nh - ---ABSENT--- - Cloud amount (low)
020012 - CL_100 - ---ABSENT--- - Cloud type (low)
020012 - CM_100 - ---ABSENT--- - Cloud type (middle)
020012 - CH_100 - ---ABSENT--- - Cloud type (high)
020013 - code - ---ABSENT--- - Height of base of lowest clouds
022012 - - ---ABSENT--- - Period of wind waves
022022 - - ---ABSENT--- - Height of wind waves
022003 - SW1 - ---ABSENT--- - Direction of 1st swell
022013 - SW1 - ---ABSENT--- - Period of 1st swell
022023 - SW1 - ---ABSENT--- - Height of 1st swell
022003 - SW2 - ---ABSENT--- - Direction of 2nd swell
022013 - SW2 - ---ABSENT--- - Period of 2nd swell
022023 - SW2 - ---ABSENT--- - Height of 2nd swell
-----
004192;OBS;1;2020/02/03 14:00:00; <-- date-heure observation
ENTETE_SHIP:YYGGw= 03144
===== CALCUL RETARD DE DIFFUSION =====
date courante : heure=14 minute=0 secondes=19 jour=3 mois=2 annee=2020 temps=1580738419
date obs      : heure=14 jour=3 mois=2 annee=2020 temps=1580738400
LAPS          : 0.005278h
entete=SNVF25 LFPW 031400
```

fig 54 -part of- format#101 decompression log

More additional format#101 log data can be found in directory: ../logs/format_101/temp/

9.9) Linux (64bit) installation

1) Copy “TurboWin+_JPMS.tar.xz” to a directory of your choice and untar the file (e.g. unpack in the newly created dir turbowin)

2) Create a shortcut to TurboWin+ JPMS, there are different ways to do this, partly dependent on the Linux distribution, see below for an Ubuntu 18 example

3) Creating application launcher on Ubuntu: You have to install gnome-panel package which comes up with ability to create an application launcher on the desktop or wherever you like.

Add --no-install-recommends suffix to prevent other package that aren't necessary.

```
sudo apt-get install --no-install-recommends gnome-panel
```

After installing gnome-panel, use the following command to create a launcher.

```
gnome-desktop-item-edit --create-new ~/Desktop
```

Once you execute the above command, create launcher application will be opened.

4) Insert Name: TurboWin+ [64-bit]

Insert Description: version x.x.x

Insert Command: PATH_TO_TURBOWIN+_BIN/java -m Turbowin_jws/turbowin.main

5) Icon

The custom TurboWin+ desktop icon can be set by clicking on the default gnome panel icon (properties) and pointing to and opening the TurboWin+ icon (“TurboWin+_install.ico”, part of the unpacked “TurboWin+_JPMS.tar.xz” file)

6) Working directory (necessary to set!)

Unfortunately the TurboWin+ working directory can't be set with the create launcher application but must be set manually:

go to the just created shortcut in ~/Desktop, open it with a text editor and add a line at the end that says Path=PATH_TO_TURBOWIN+_BIN (see example below)

Example (text) gnome panel start up TurboWin+ JPMS launcher (dir ~/Desktop):

```
[Desktop Entry]
Version=1.0
Type=Application
Terminal=false
Icon[en_US]=gnome-panel-launcher
Name[en_US]=TurboWin+ [64-bit]
Comment[en_US]=version 4.0
Exec=/home/martin/turbowin/TurboWin+_JPMS/turbowin_jws/bin/java -m Turbowin_jws/turbowin.main
Path=/home/martin/turbowin/TurboWin+_JPMS/turbowin_jws/bin
Name=TurboWin+ [64-bit]
Comment=version 4.0
Icon=/home/martin/turbowin/TurboWin+_install.ico
```

7) USB

If an instrument will be connected to TurboWin+ you must have full and direct access to the USB ports, to set these permissions (once):

```
sudo adduser <the user you want to add> dialout
```

```
sudo reboot
```

8) For troubleshooting you can start TurboWin+ with the launcher to monitor all the incoming and outgoing messages,

go to the appropriate directory and start *./turbowin_launcher*

note Ubuntu 19 and 20

It seems gnome-desktop-item-edit is removed from the gnome-panel package, it won't be available for Ubuntu 19.10 (and later releases) as a workaround you can create .desktop files by hand. Just create a blank text file and add the appropriate lines to it (e.g. see above). Then rename the file as e.g. turbowin.desktop and right-click and select 'allow launching' (you may also have to make it executable).

NB Many .desktop files for installed applications are usually found in your /usr/share/applications/ directory

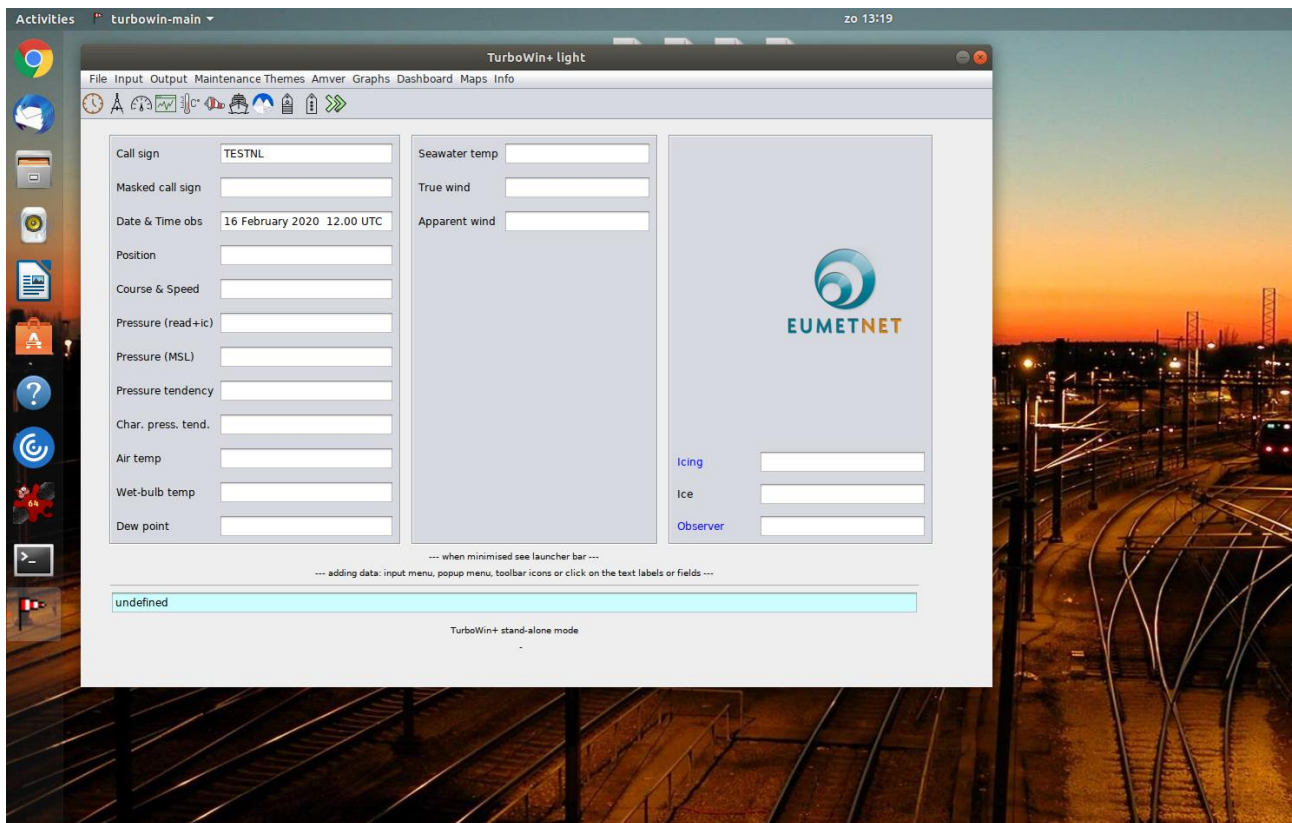


fig 55 TurboWin+ on Ubuntu 18.04 LTS

9.10) Obs to server (Rest API)

TurboWin+ Server-side API

In TurboWin+ there are options to send (upload) a meteorological observation straight to the National Meteorological Service or to a Centralized Service. This is the fastest way to get an observation from the ship into the GTS. There are slight differences in the server-side API implementation of FM13 and format #101 observations.

Format #101 API

In TurboWin+ (from version 2.5.1) there is the option to send the format #101 observation straight to a server of a National Meteorological Service or Centralized Service.

Upload URL format 101:

TurboWin+ sends a GET HTTP request to the URL which can be set in TurboWin+.

The server must expect the HTTP request, in the form of GET. When received, the server must interpret and validate the information supplied and respond as below

upload URL which can be set manually as a setting in TurboWin+: --- consult your PMO ---

So the server site script could be placed on every server with every path and could be named "index_webstart_101.php" but also this is not mandatory. But of course the full URL must be set once manually in TurboWin+ and must always end with a "?"

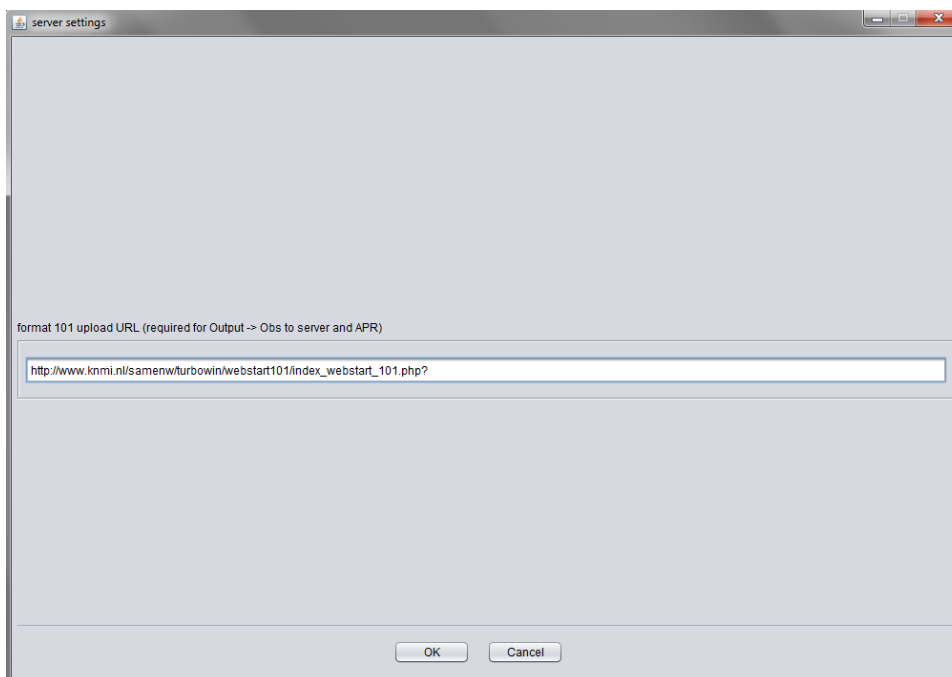


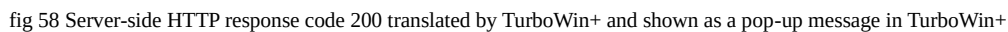
fig 56 TurboWin+ input screen to set upload URL (inserted URL as an example)

TurboWin+ will add the weather observation as key/value pair indicating the weather data. Key: "obs" the weather data itself is according format#101 and also encoded by TurboWin+. TurboWin+

Example of a full upload string:

note: So on the server the received observation string must be decoded e.g. with php function `urldecode` (Decodes any `###` encoding in the given string. Plus symbols ('+') are decoded to a space character)

All requests must return a HTTP response (status) code. A success is indicated by 200. Anything else is a failure.



In TurboWin+ it is also possible to send a FM13 observation straight to a server of a National Meteorological Service or Centralized Service.

The information in this chapter is subject to change. For the latest info please send a mail to pmo.office@knmi.nl

On request the php scrips of the KNMI server-side implementation can be obtained from pmo.office@knmi.nl

*Ethernet						
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
Apply a display filter: <Ctrl>F						
No.	Time	Source	Destination	Protocol	Length	Info
43	12.000467	Tp-LinkT_d1:3c:eb	Spanning-tree-for...	STP	60	Conf. Root = 32768/0/ec:08:6b:d1:3c:ec Cost = 0 Port = 0x8003
44	12.462136	192.168.178.59	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.252 for any sources
45	12.582891	145.23.3.62	192.168.178.45	TLSv1.2	81	Application Data
46	12.625138	192.168.178.45	145.23.3.62	TCP	54	64879 → 443 [ACK] Seq=1 Ack=136 Win=1026 Len=0
47	12.874197	192.168.178.234	255.255.255.255	UDP	214	45836 → 6667 Len=172
48	12.963916	192.168.178.45	142.250.102.188	TCP	55	59333 → 5228 [ACK] Seq=1 Ack=1 Win=1025 Len=1
49	12.980060	142.250.102.188	192.168.178.45	TCP	66	5228 → 59333 [ACK] Seq=1 Ack=2 Win=265 Len=0 SLE=1 SRE=2
50	13.166028	192.168.178.45	89.101.251.228	DNS	92	Standard query 0x10bb A turbowin-rest-api.pmc.knmi.cloud
51	13.195043	89.101.251.228	192.168.178.45	DNS	140	Standard query response 0x10bb A turbowin-rest-api.pmc.knmi.cloud A 34.240.208.120 A 34.240.156.249 A 52.210.10.209
52	13.196055	192.168.178.45	34.240.208.120	TCP	66	56893 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
53	13.229021	34.240.208.120	192.168.178.45	TCP	66	443 → 56893 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1420 SACK_PERM=1 WS=256
54	13.229137	192.168.178.45	34.240.208.120	TCP	54	56893 → 443 [ACK] Seq=1 Ack=1 Win=262656 Len=0
55	13.243985	192.168.178.45	34.240.208.120	TLSv1.2	346	Client Hello
56	13.277755	34.240.208.120	192.168.178.45	TCP	60	443 → 56893 [ACK] Seq=1 Ack=293 Win=30464 Len=0
57	13.279030	34.240.208.120	192.168.178.45	TLSv1.2	1474	Server Hello
58	13.280123	34.240.208.120	192.168.178.45	TCP	1474	443 → 56893 [ACK] Seq=1421 Ack=293 Win=30464 Len=1420 [TCP segment of a reassembled PDU]
59	13.280159	192.168.178.45	34.240.208.120	TCP	54	56893 → 443 [ACK] Seq=293 Ack=2841 Win=262656 Len=0
60	13.280616	34.240.208.120	192.168.178.45	TCP	1474	443 → 56893 [ACK] Seq=2841 Ack=293 Win=30464 Len=1420 [TCP segment of a reassembled PDU]
61	13.280616	34.240.208.120	192.168.178.45	TLSv1.2	1105	Certificate, Server Key Exchange, Server Hello Done
62	13.280655	192.168.178.45	34.240.208.120	TCP	54	56893 → 443 [ACK] Seq=293 Ack=5312 Win=262656 Len=0
63	13.290454	192.168.178.45	34.240.208.120	TLSv1.2	129	Client Key Exchange
64	13.298682	192.168.178.45	34.240.208.120	TLSv1.2	60	Change Cipher Spec
65	13.298885	192.168.178.45	34.240.208.120	TLSv1.2	99	Encrypted Handshake Message
66	13.329433	34.240.208.120	192.168.178.45	TCP	60	443 → 56893 [ACK] Seq=5312 Ack=419 Win=30464 Len=0
67	13.329433	34.240.208.120	192.168.178.45	TLSv1.2	105	Change Cipher Spec, Encrypted Handshake Message
68	13.330920	192.168.178.45	34.240.208.120	TLSv1.2	420	Application Data
69	13.402431	34.240.208.120	192.168.178.45	TCP	60	443 → 56893 [ACK] Seq=5363 Ack=793 Win=31488 Len=0
70	13.585953	192.168.178.45	3.127.179.255	TLSv1.2	100	Application Data
71	13.609341	3.127.179.255	192.168.178.45	TCP	60	443 → 52521 [ACK] Seq=727 Ack=168 Win=19 Len=0
72	13.609641	3.127.179.255	192.168.178.45	TLSv1.2	111	Application Data
73	13.663872	192.168.178.45	3.127.179.255	TCP	54	52521 → 443 [ACK] Seq=168 Ack=784 Win=1025 Len=0
74	13.748688	192.168.178.178	255.255.255.255	UDP	215	33269 → 7437 Len=173

fig 59 example network monitoring trace TurboWin+ format #101 obs upload + response

9.11 SRM (Serial communication Recovery Mode)

If an application has lost the serial communication with a device, most of the times you have to manually restart the system. Thanks to the in TurboWin+ integrated SRM (based on threading) most times it will recover automatically after loosing the serial connection. You could test this eg by removing the (serial)USB cable and after a while reconnecting the USB cable again. (Every hour [hh+43] minutes the serial connection will be tested and re-established if necessary)

```
04-Nov-2019 08:40:06 UTC [GPS] data listener thread cancelling
04-Nov-2019 08:43:06 UTC [GPS] starting-com-port closed (User-Specified Port)
04-Nov-2019 08:43:11 UTC [GPS] found serial com port (ok): Prolific USB-to-Serial Comm Port (COM12)
04-Nov-2019 08:43:15 UTC [GPS] found GPS on port: Prolific USB-to-Serial Comm Port (COM12)
04-Nov-2019 08:43:16 UTC [GPS] data listener thread restarting
04-Nov-2019 08:44:06 UTC [GPS] start receiving data again
```

fig 60 SRM logged

9.12 Check your observations

If internet available, you can check your sent observations on the E-SURFMAR VOS Control Panel (until a few days ago)

TurboWin+ select: Info → Statistics (internet).

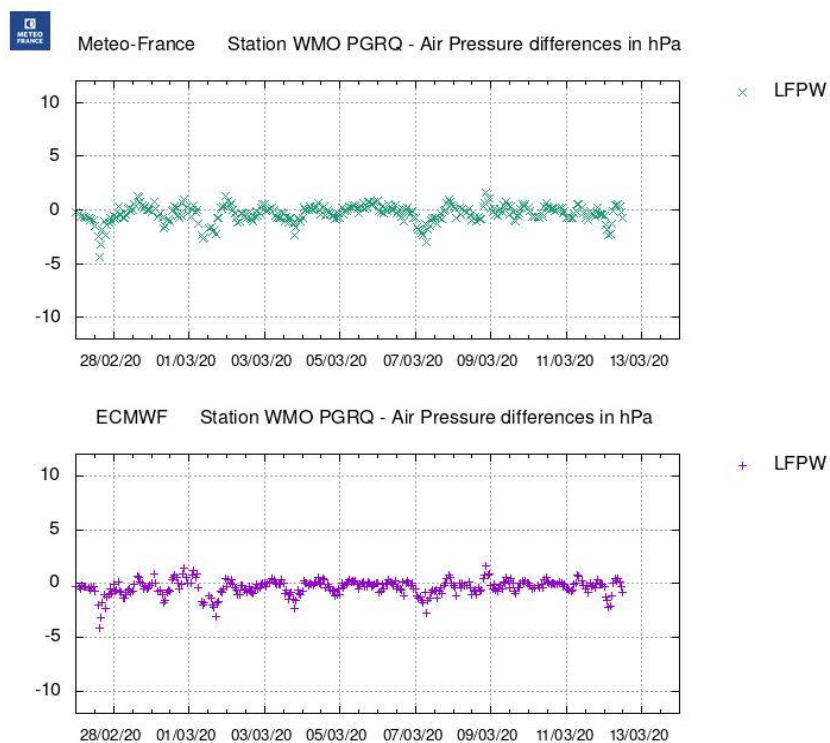


fig 61 example of air pressure quality control on the E-Surfmar VOS Control Panel

9.13a Autostart after reboot

After TurboWin+ installation you can instruct your OS to autostart the application after a OS reboot

On a computer with Windows OS:

- 1) Press the Windows key + R to open the Run box. Type **shell:startup** and press Enter. (It will reach the start-up folder)
- 2) Copy the TurboWin+ desktop icon (right click) and paste into the start-up folder. (or Right click in the Start-up folder for pop-up menu: New → Shortcut → Browse... → Select TurboWin+ → OK → Next → Finish)
- 3) Restart the computer to test it
- 4) Note, with the W10 settings menu you also can verify TurboWin+ autostart [fig 62 and fig 63]

9.13b Autostart after reboot and after sign in

If officers must log into their own account/virtual machine when they come on watch it is also possible to automatically start-up TurboWin+ after every log in.

On a computer with Windows OS:

- 1] "Turn On Automatically Restart Apps after Sign In": via the main (administrator) account: Settings -> Accounts -> sign-in- options -> Restart apps: ON (Available from at least Windows 10 build 18963)
- 2] for every account:
 - a) Press the Windows key + R to open the Run box. Type **shell:startup** and press Enter. (It will reach the start-up folder)
 - b) Copy the TurboWin+ desktop icon (right click) and paste into the start-up folder. (or Right click in the Start-up folder for pop-up menu: New → Shortcut → Browse... → Select TurboWin+ → OK → Next → Finish)
- 3) Sign out with the present account and sign in with a second account to test it
- 4) Note, with the W10 settings menu you also can verify TurboWin+ autostart [fig 62 and fig 63]

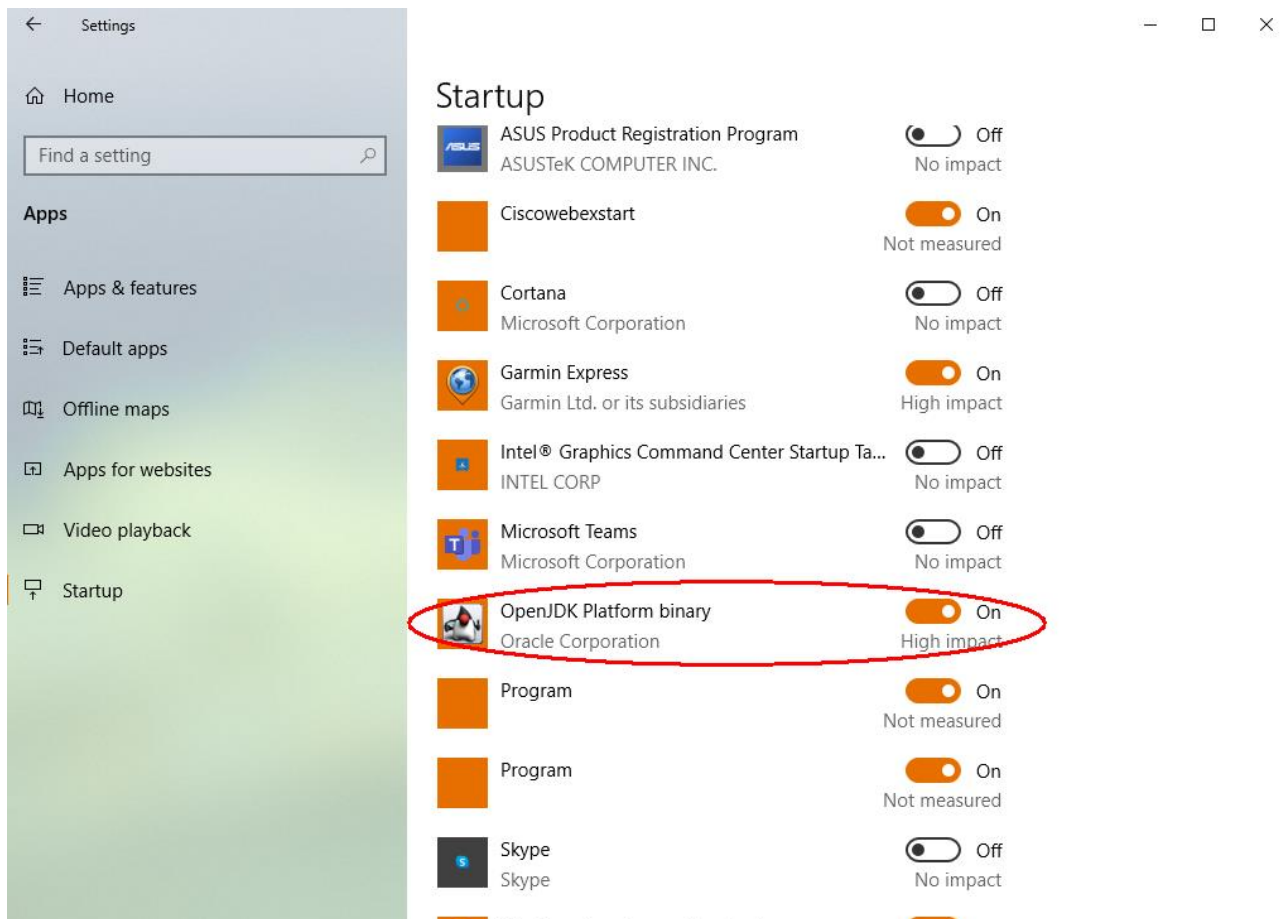


fig 62 Windows 10 64bit startup setting for TurboWin+ autostart

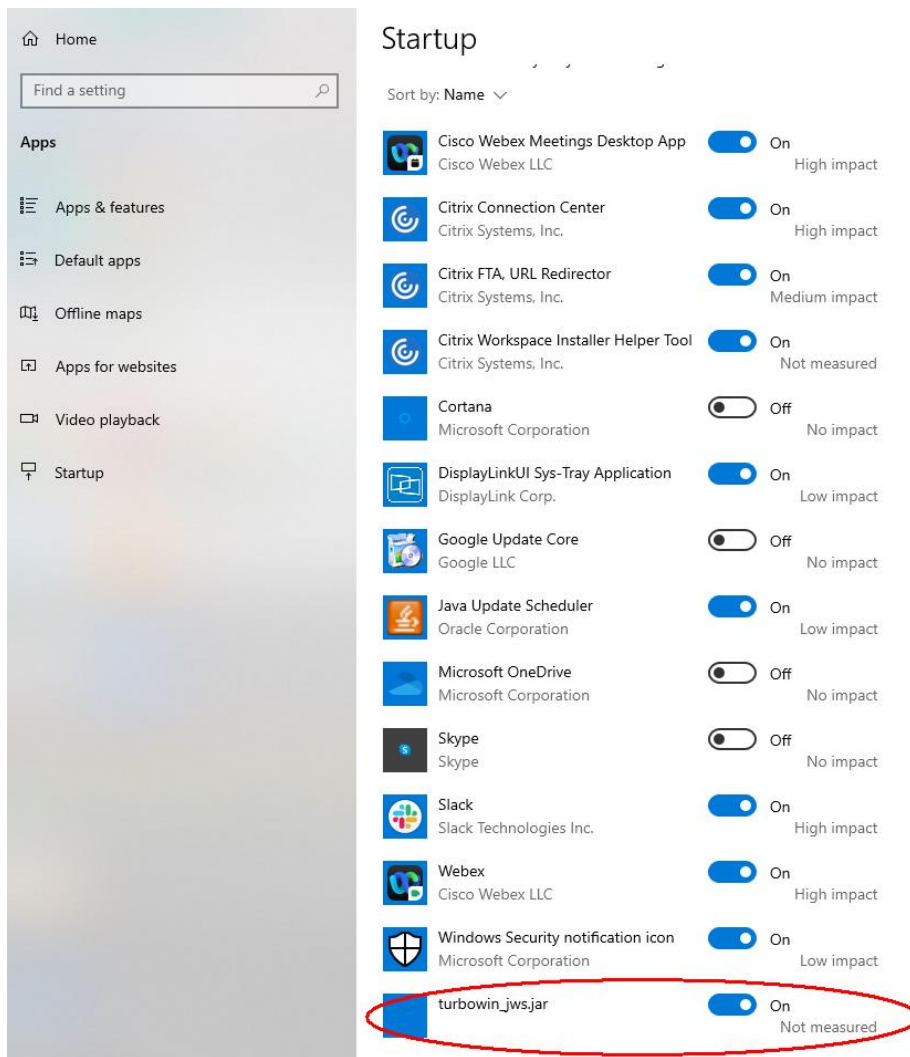


fig 63 Windows 10 32bit startup setting for TurboWin+ autostart